

Optics and Modern Physics

CHAPTER 9

Communication Systems

9.1–9.10

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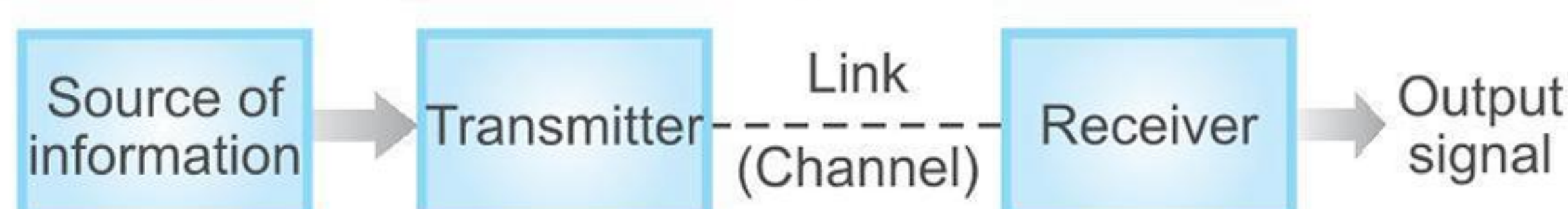
Communication Systems

INTRODUCTION

The term communication refers to the transmitting, receiving, and processing of information by electronic means.

BASIC COMMUNICATION SYSTEM

A basic communication system consists of an information source, a transmitter, a link, and a receiver.



Information: The idea/message that is to be conveyed is information. The message may be individual one or a set of messages. It may be a symbol, code, group of words, or any predecided unit.

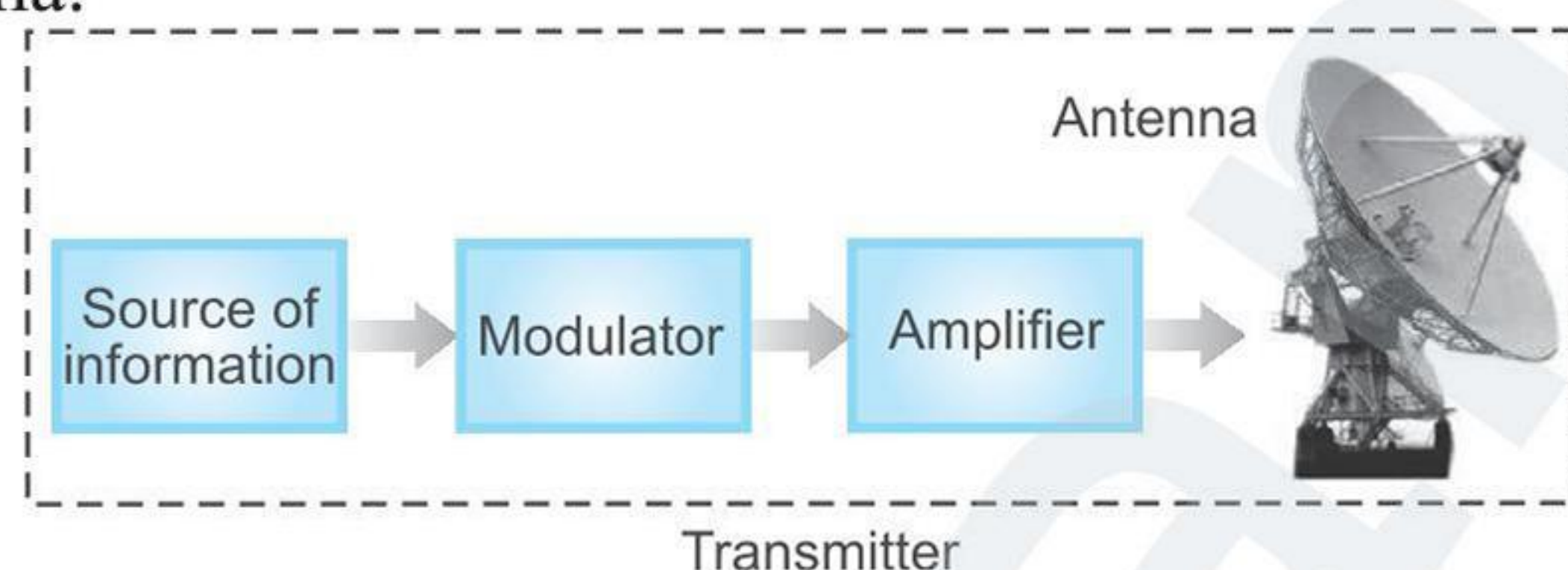
Transmitter: In radio transmission, the transmitter consists of a transducer, modulator, amplifier, and transmitting antenna.

Transducer: Converts sound signals into electric signal.

Modulator: Mixing of audio electric signal with high frequency radio wave.

Amplifier: Boosting the power of modulated signal.

Antenna: Signal is radiated in the space with the aid of an antenna.



Communication channel: The function of communication channel is to carry the modulated signal from transmitter to receiver. The communication channel is also called transmission medium or link.

The term channel refers to the frequency range allocated to a particular service or transmission.

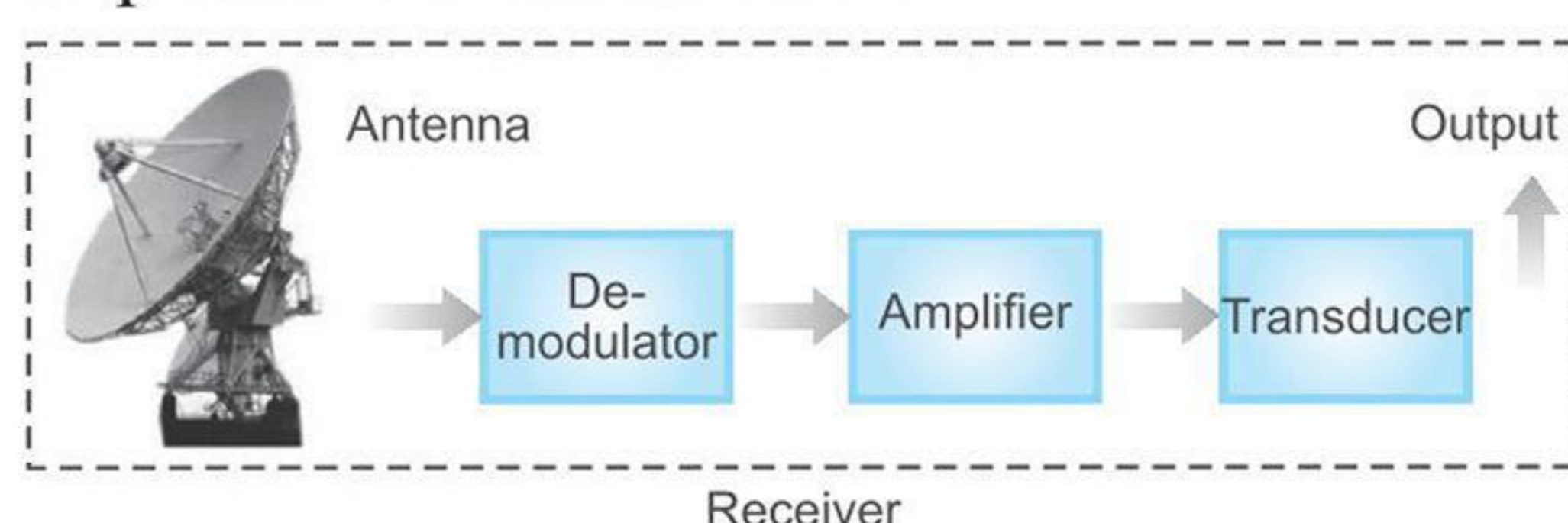
Receiver: A receiver consists of the following components:

Pickup antenna: To pick the signal

Demodulator: To separate out the audio signal from the modulated signal

Amplifier: To boost up the weak audio signal

Transducer: To convert back audio signal in the form of electrical pulses into sound waves.



TYPES OF COMMUNICATION SYSTEM

Communication systems can be classified according to the nature of information or mode of transmission or types of transmission channel or types of modulation.

Classification according to the nature of information source:

1. Speech transmission
2. Picture transmission
3. Facsimile transmission (FAX): This involves exact reproduction of a document or picture which are static.

Classification according to the mode of transmission:

1. **Analog communication:** The communication system which makes use of analog signals is called analog communication system.
2. **Digital communication:** In this system, digital signals are used.

Classification according to the transmission channel:

1. Line communication
2. Space communication

Classification according to the type of modulation:

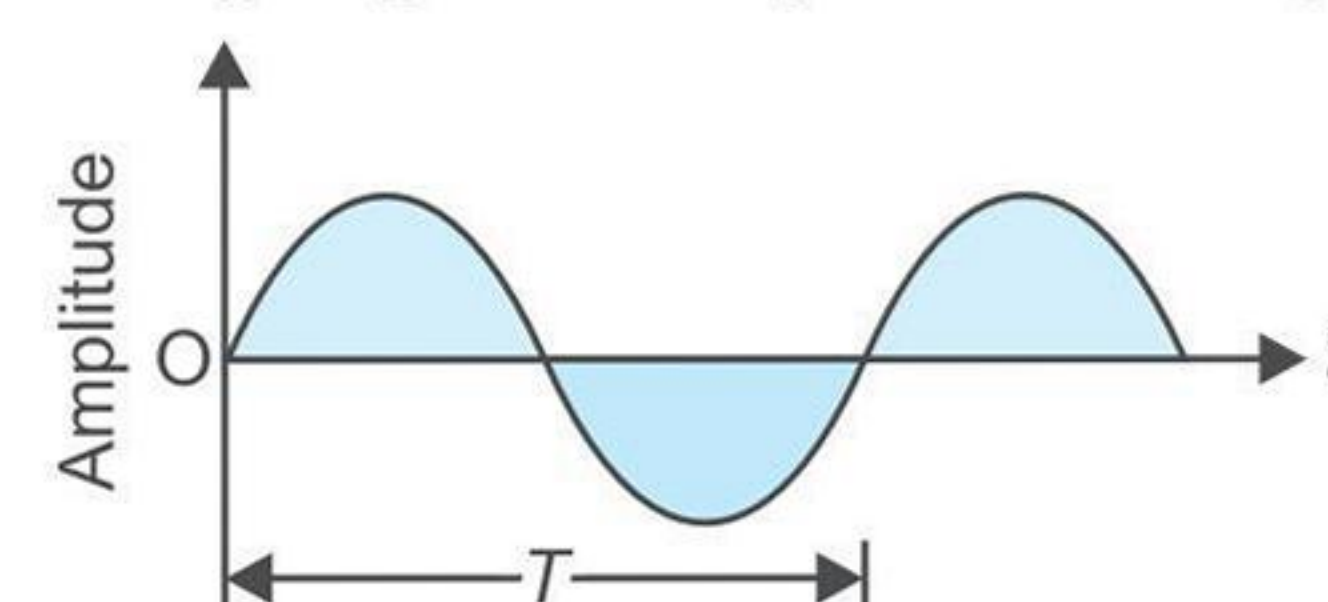
1. Amplitude modulation (AM)
2. Frequency modulation (FM)
3. Phase modulation (PM)
4. Pulse amplitude modulation (PAM)
5. Pulse time modulation (PTM)
6. Pulse code modulation (PCM)

ANALOG AND DIGITAL SIGNALS

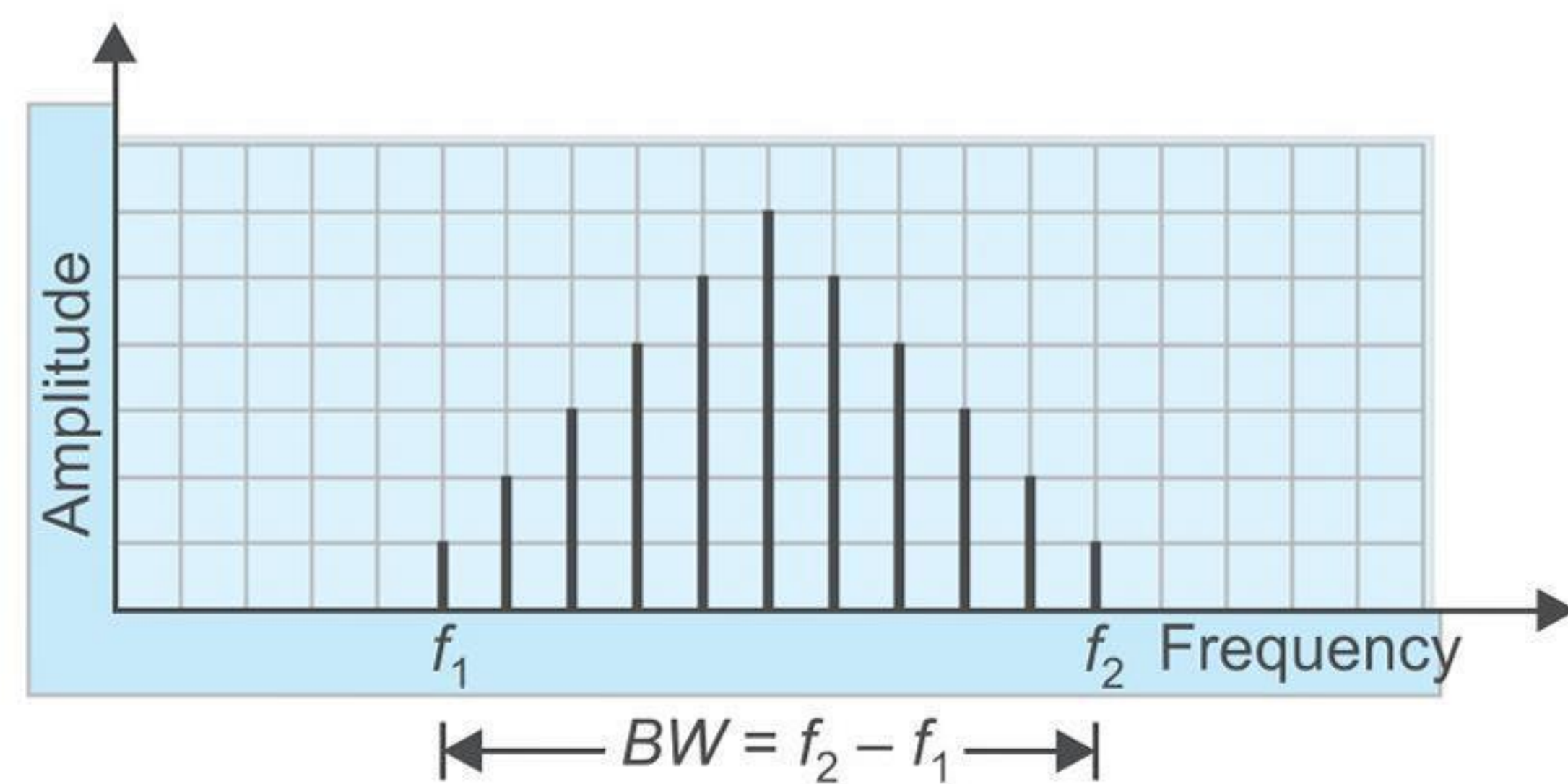
In communication system, a signal means a time, varying electrical signal containing informations.

Analog signals: It is a continuous wave form which changes smoothly over time.

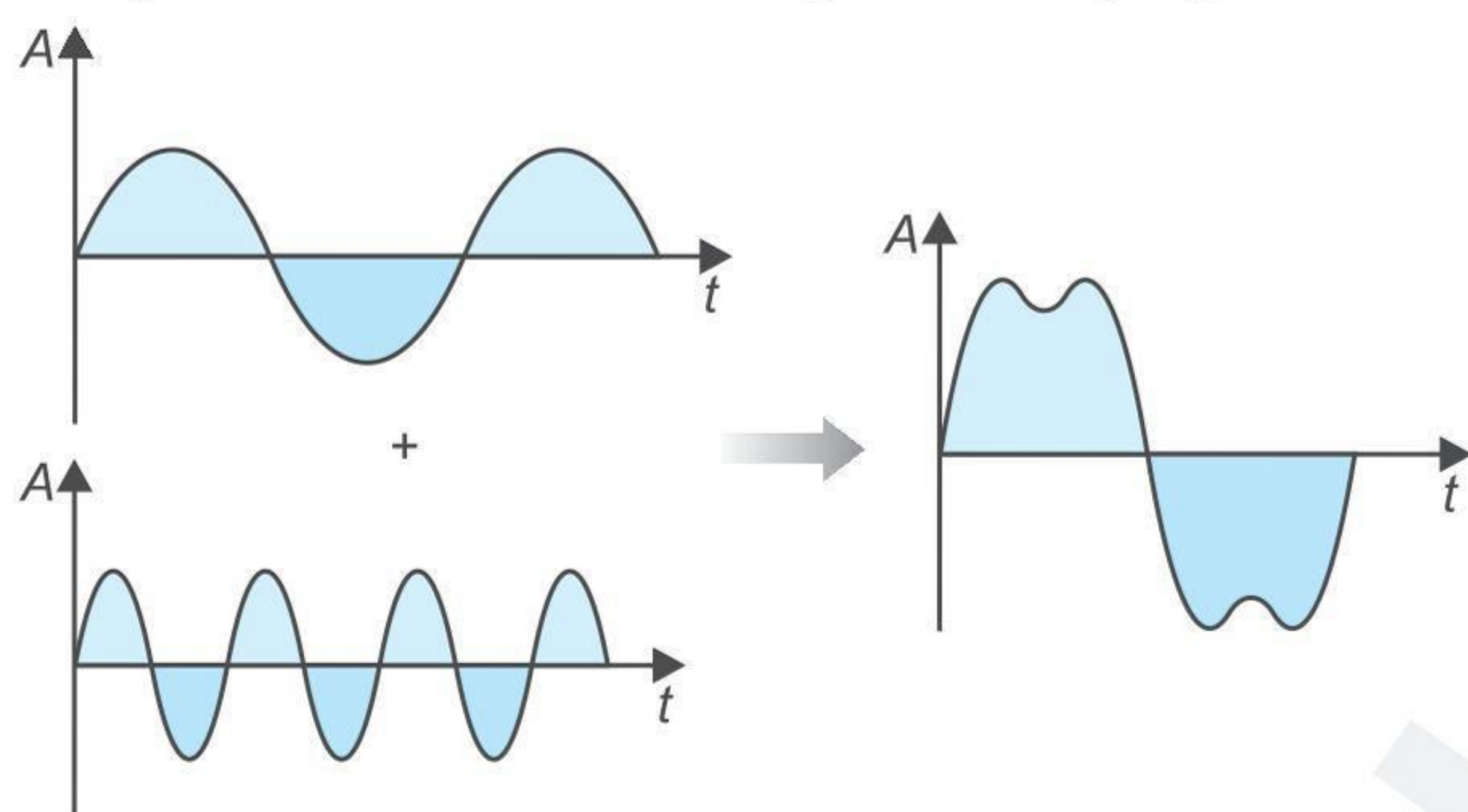
- Such signals can be easily generated from the source of information by using an appropriate transducer, e.g., pressure variations in the sound waves can be converted into corresponding current or voltage pulses with the help of a microphone.
- A simple analog signal is represented by a sine wave.



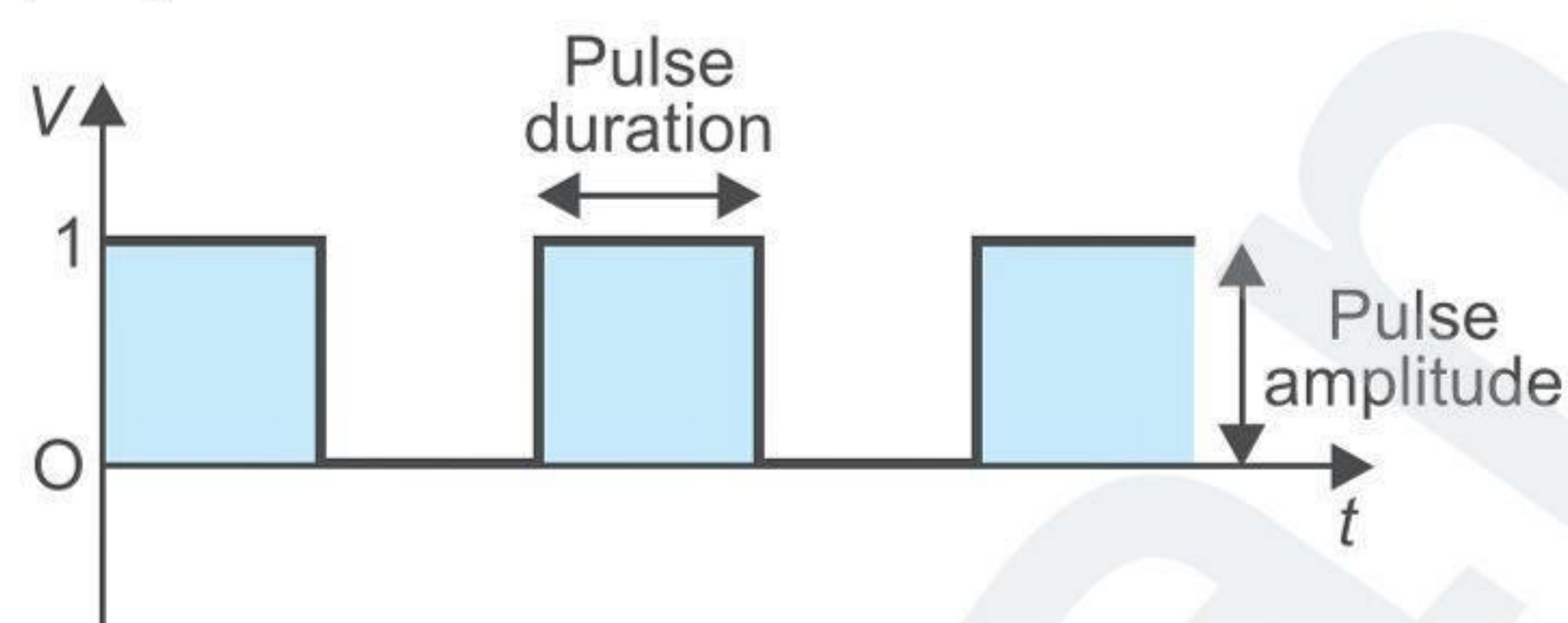
- The frequency of analog signals associated with speed or music varies over a range between 20 Hz to 20 kHz.



- The range over which the frequencies of a signal vary is called bandwidth.
- The term base band designates the band of frequencies representing the signal supplied by the source of information.
- A signal consists of two or more waves of different frequencies known as a complex analog signal.



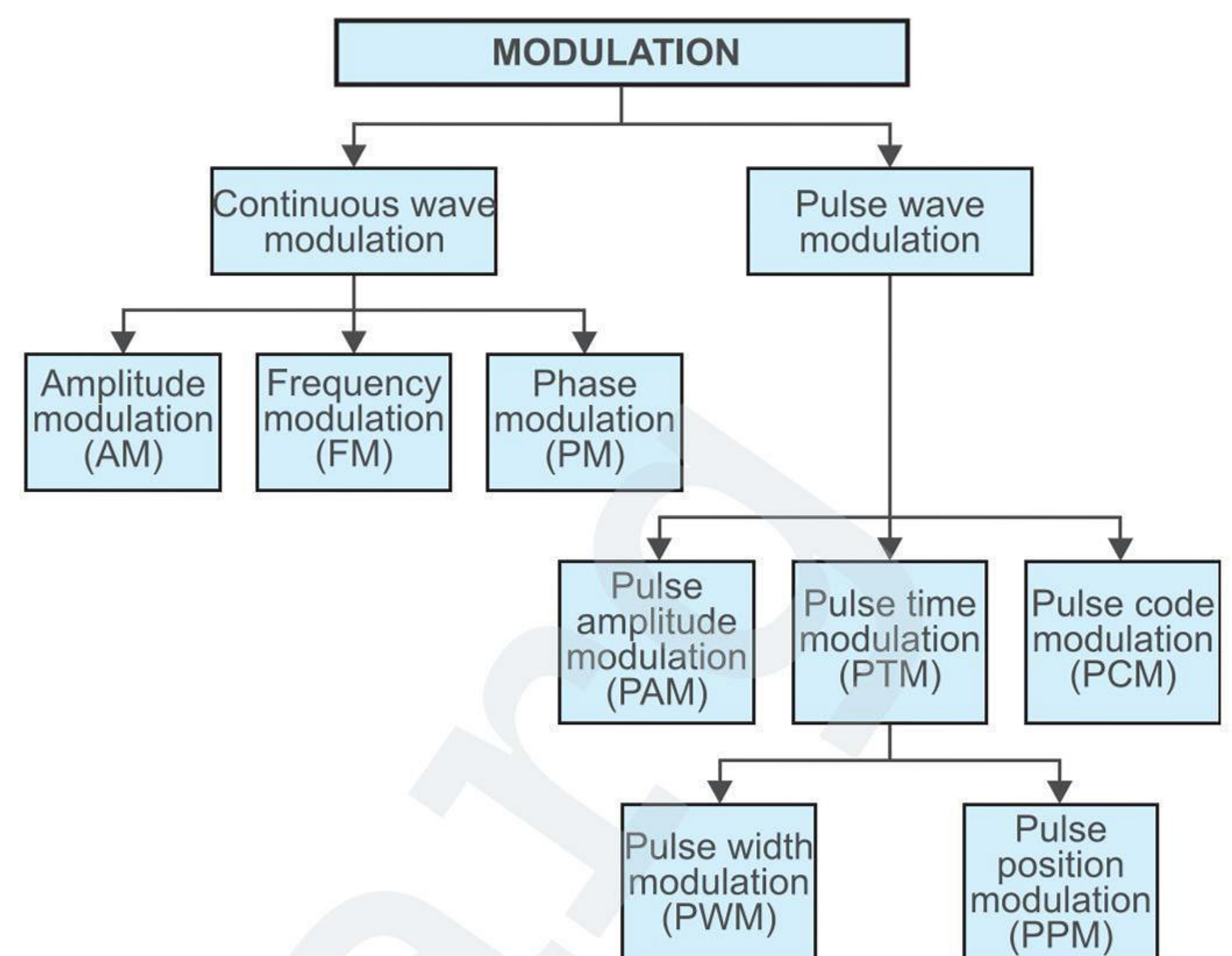
Digital signals: A digital signal is a discontinuous function of time. It has only two voltage levels, i.e., either low (0) or high (1). Either of 0 and 1 is known as bit. A group of bit is called byte. A byte comprising 2 bits can give four code combinations, i.e., 00, 01, 10, and 11.



The number of code combination increases with number of bits in a byte is given by $N = 2^x$, where x = number of bits in a byte. The number of binary digits (bits) per second, which describe a digital signal is called its bit rate. Bit rate is expressed in bits per second (bps).

MODULATION

Digital and analog signals to be transmitted are usually of low frequency and hence cannot be transmitted as such. These signals require some carrier to be transported. These carriers are known as carrier waves or high-frequency signals. The process of placement of a low-frequency (LF) signal over the high-frequency (HF) signal is known as modulation. Various types of modulations are shown in figure.



NEED FOR MODULATION

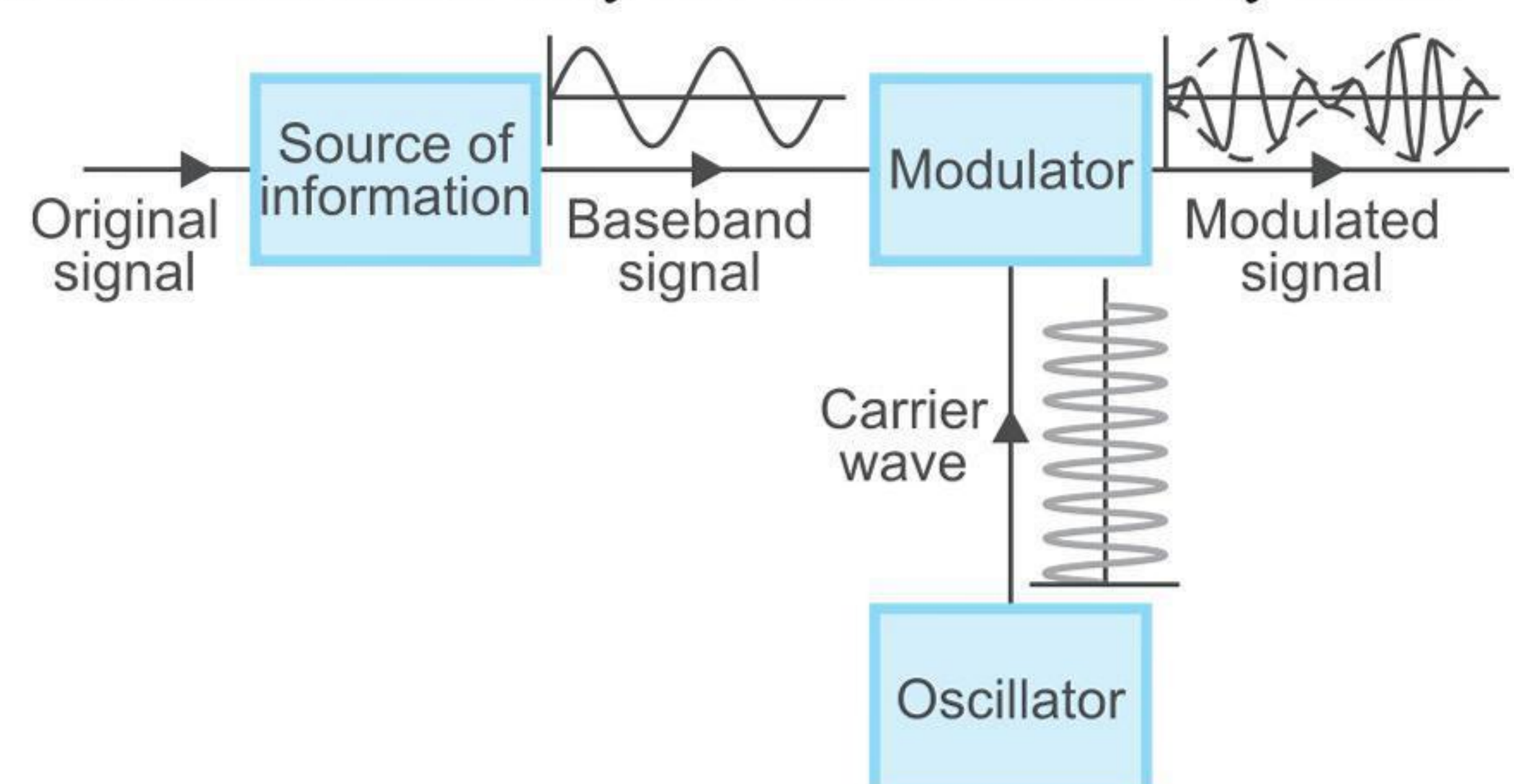
A sound wave (20 Hz to 20 kHz) cannot be transmitted directly from one place to another for the following reasons.

Height of Antenna

For efficient radiation and reception, the height of transmitting and receiving antennas should be comparable to a quarter of wavelength of the frequency used. For 15 kHz it is 5000 m (too large) and for 1 MHz it is 75 m. The energy radiated from an antenna is practically zero, when the frequency of the signal to be transmitted is below 15 Hz.

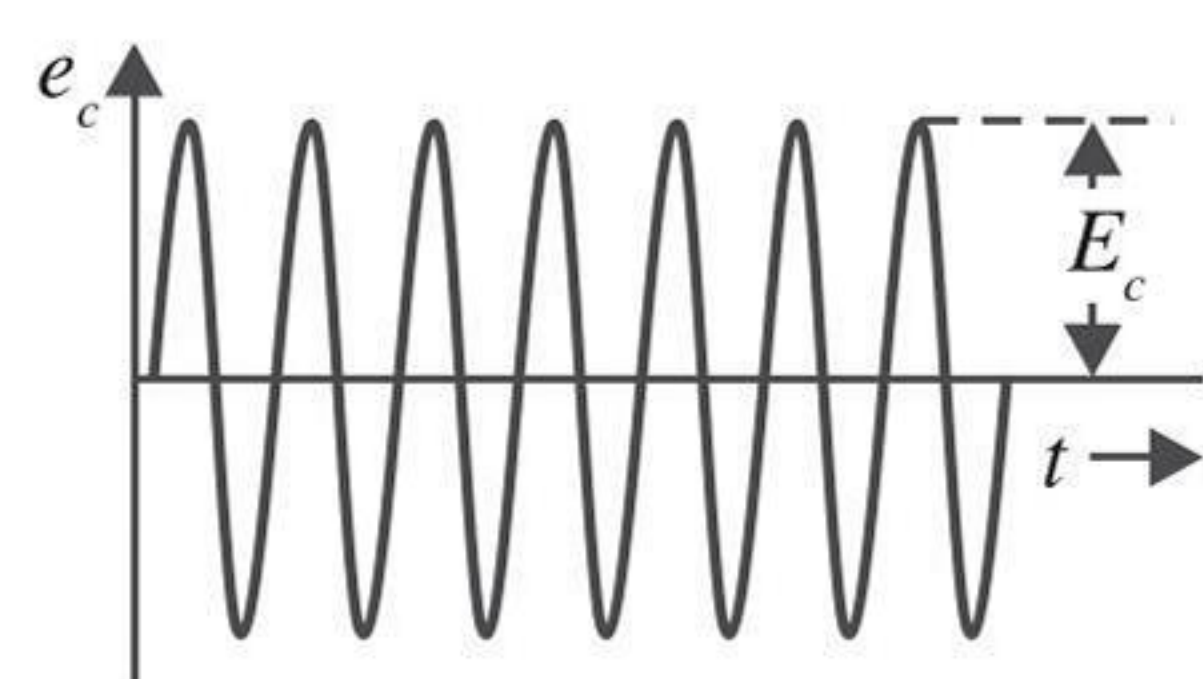
Detecting Signals

All audible signals are in the range of 20 Hz to 20 kHz. So the signals from all sources remain heavily mixed up in air. It will be very difficult to differentiate or detect the broadcast signal at the receiving station. Thus, modulation is necessary for a low-frequency signal especially when it is to be sent to a distant place, so that the information may not die out in the way itself.

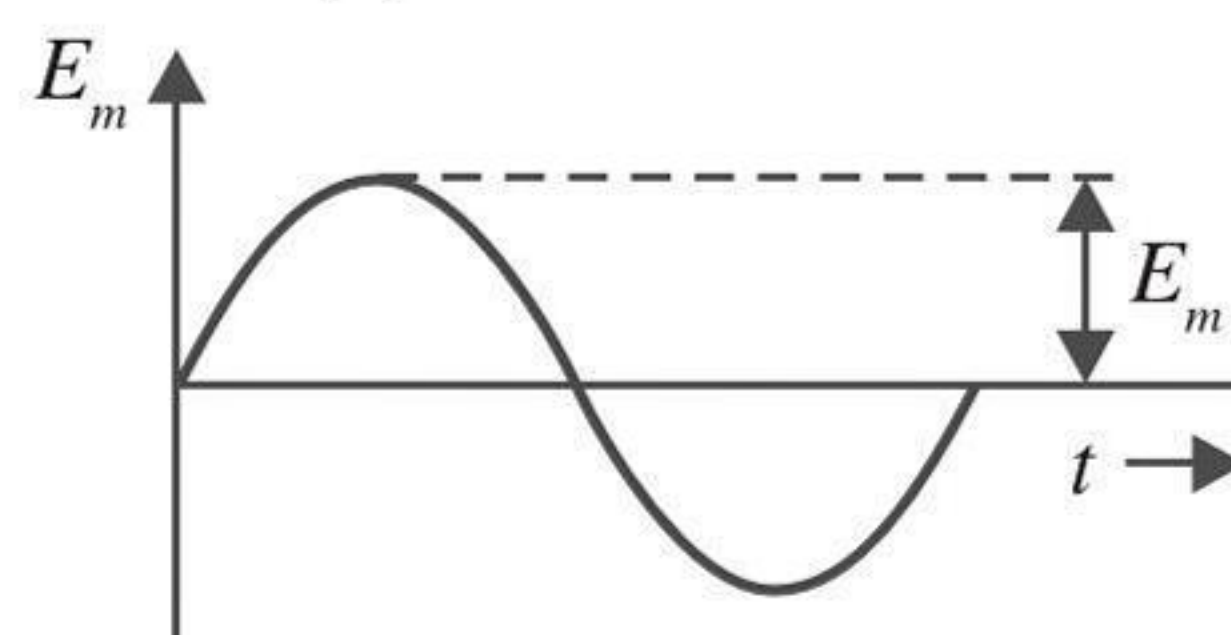


AMPLITUDE MODULATION

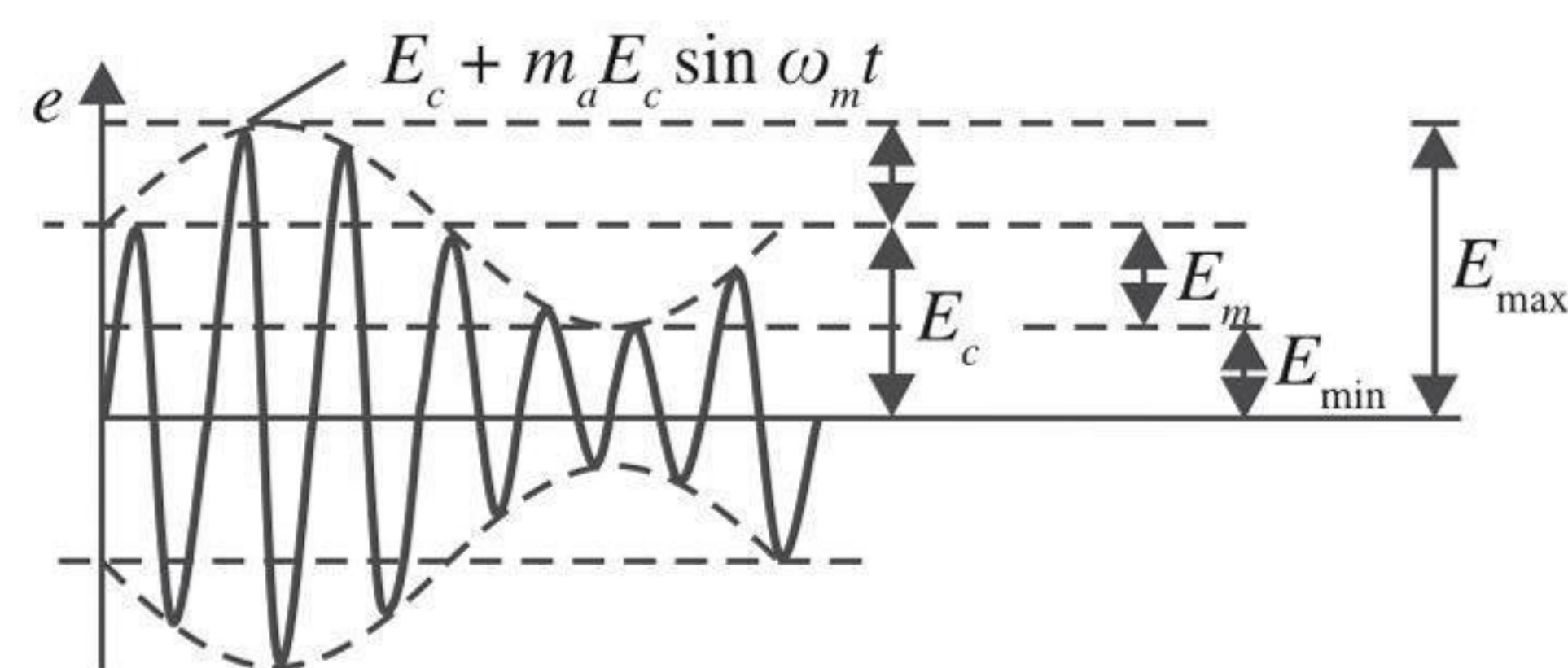
The process of changing the amplitude of a carrier wave in accordance with the amplitude of the audio frequency (AF) signal is known as amplitude modulation (AM). In AM frequency of the carrier wave remains unchanged. The amplitude of modulated wave is varied in accordance with the amplitude of modulating wave.



(b) HF carrier wave



(b) LF modulating



(c) AM wave

Modulation index: The ratio of change of amplitude of the carrier wave to the amplitude of original carrier wave is called the modulation factor or degree of modulation or modulation index (m_a).

$$m_a = \frac{\text{Change in amplitude of carrier wave}}{\text{Amplitude of original carrier wave}} = \frac{kE_m}{E_c}$$

where k is a factor which determines the maximum change in the amplitude for a given amplitude E_m of the modulating signal. If $k = 1$, then

$$m_a = \frac{E_m}{E_c} = \frac{E_{\max} - E_{\min}}{E_{\max} + E_{\min}}$$

If a carrier wave is modulated by several sine waves, the total modulated index m_t is given by

$$m_t = \sqrt{m_1^2 + m_2^2 + m_3^2 + \dots}$$

Voltage equation for AM wave: Suppose voltage equations for carrier wave and modulating wave are $e_c = E_c \cos \omega_c t$ and $e_m = E_m \sin \omega_m t = m_a E_c \sin \omega_m t$, where e_c is the instantaneous voltage of carrier wave, E_c amplitude of carrier wave, $\omega_c (=2\pi f_c)$ angular velocity at carrier frequency f_c , e_m instantaneous voltage of modulating, E_m amplitude of modulating wave, and $\omega_m (=2\pi f_m)$ angular velocity of modulating frequency f_m .

Voltage equation for AM wave is

$$\begin{aligned} e &= E \sin \omega_c t \\ &= (E_c + e_m) \sin \omega_c t \\ &= (E_c + E_m \sin \omega_m t) \sin \omega_c t \\ &= E_c \sin \omega_c t + \frac{m_a E_c}{2} \cos(\omega_c - \omega_m)t - \frac{m_a E_c}{2} \cos(\omega_c + \omega_m)t \end{aligned}$$

The above AM wave indicated that the AM wave is equivalent to the summation of three sinusoidal waves, one having amplitude E_c and the other two having amplitude $m_a E_c / 2$.

Side band frequencies: An AM wave contains three frequencies: f_c (called carrier frequency), $(f_c + f_m)$, and $(f_c - f_m)$ (latter two called

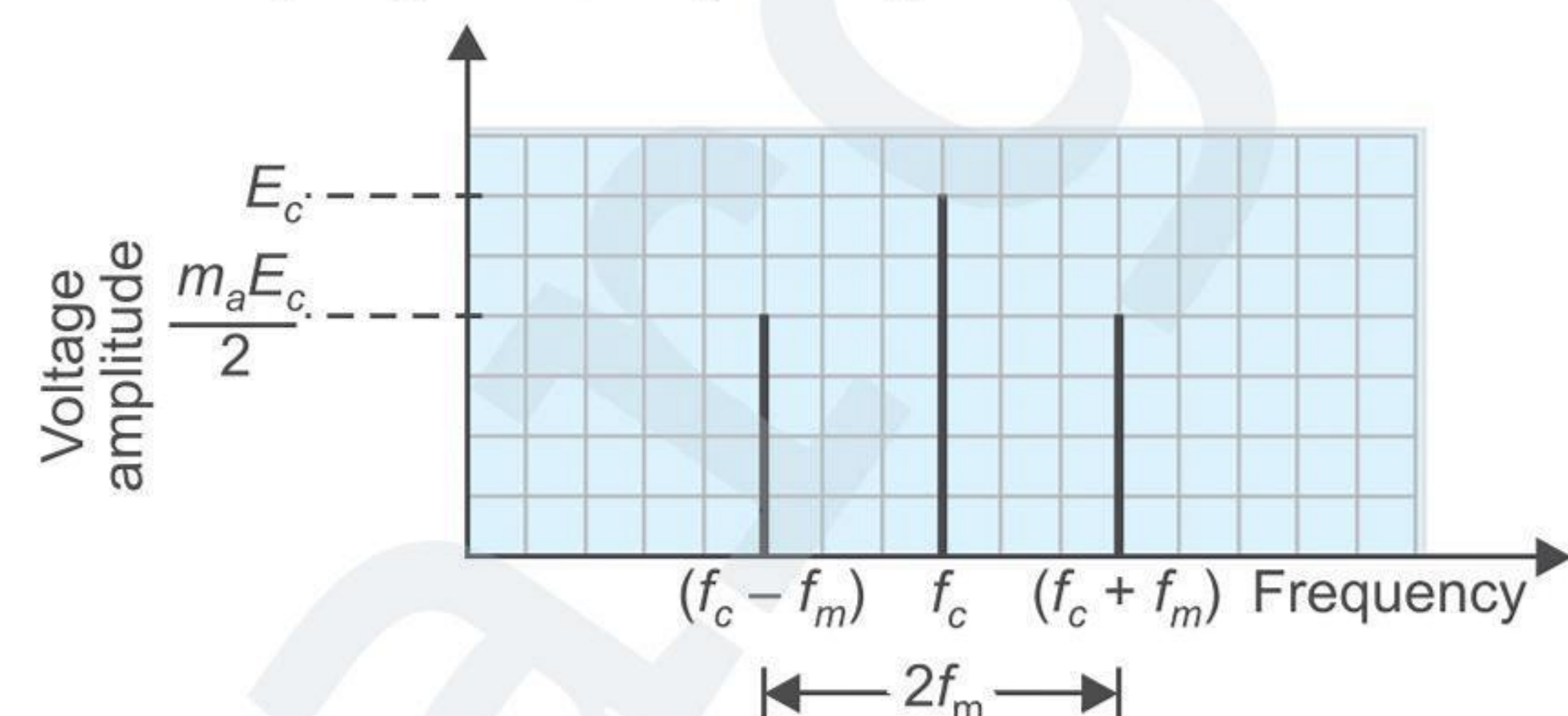
side band frequencies).

$(f_c + f_m)$: Upper side band (USB) frequency

$(f_c - f_m)$: Lower side band (LSB) frequency

Side band frequencies are generally close to the carrier frequency.

Bandwidth in AM wave: The two side bands lie on either side of the carrier frequency at equal frequency interval f_m . So, bandwidth = $(f_c + f_m) - (f_c - f_m) = 2f_m$.



Power in AM waves: Power dissipated in any circuit,

$P = V_{\text{rms}}^2 / R$. Hence,

$$\text{Carrier power, } P_c = \frac{(E_c / \sqrt{2})^2}{R} = \frac{E_c^2}{2R}$$

Total power of side bands,

$$P_{sb} = \frac{(m_a E_c / 2\sqrt{2})^2}{R} + \frac{(m_a E_c / 2\sqrt{2})^2}{R} = \frac{m_a^2 E_c^2}{4R}$$

Total power of AM wave, $P_t = P_c + P_{sb} = \frac{E_c^2}{2R} \left(1 + \frac{m_a^2}{2} \right)$

$$\frac{P_t}{P_c} = \left(1 + \frac{m_a^2}{2} \right) \text{ and } \frac{P_{sb}}{P_t} = \frac{m_a^2 / 2}{\left(1 + \frac{m_a^2}{2} \right)}$$

Maximum power in AM (without distortion) will occur when $m_a = 1$, i.e., $P_t = 1.5P_c = 3P_{sb}$

If I_c = unmodulated current and I_t = total or modulated current,

$$\text{then } \frac{P_t}{P_c} = \frac{I_t^2}{I_c^2} \Rightarrow \frac{I_t}{I_c} = \sqrt{\left(1 + \frac{m_a^2}{2} \right)}$$

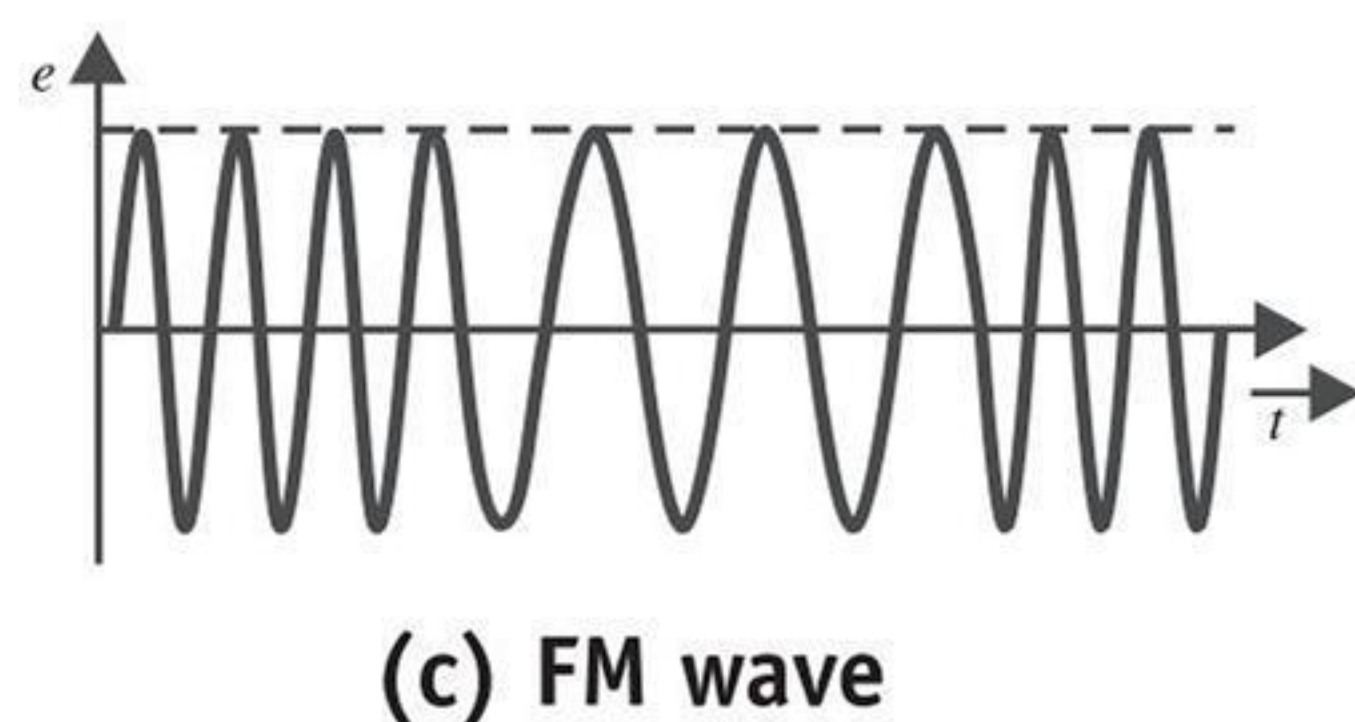
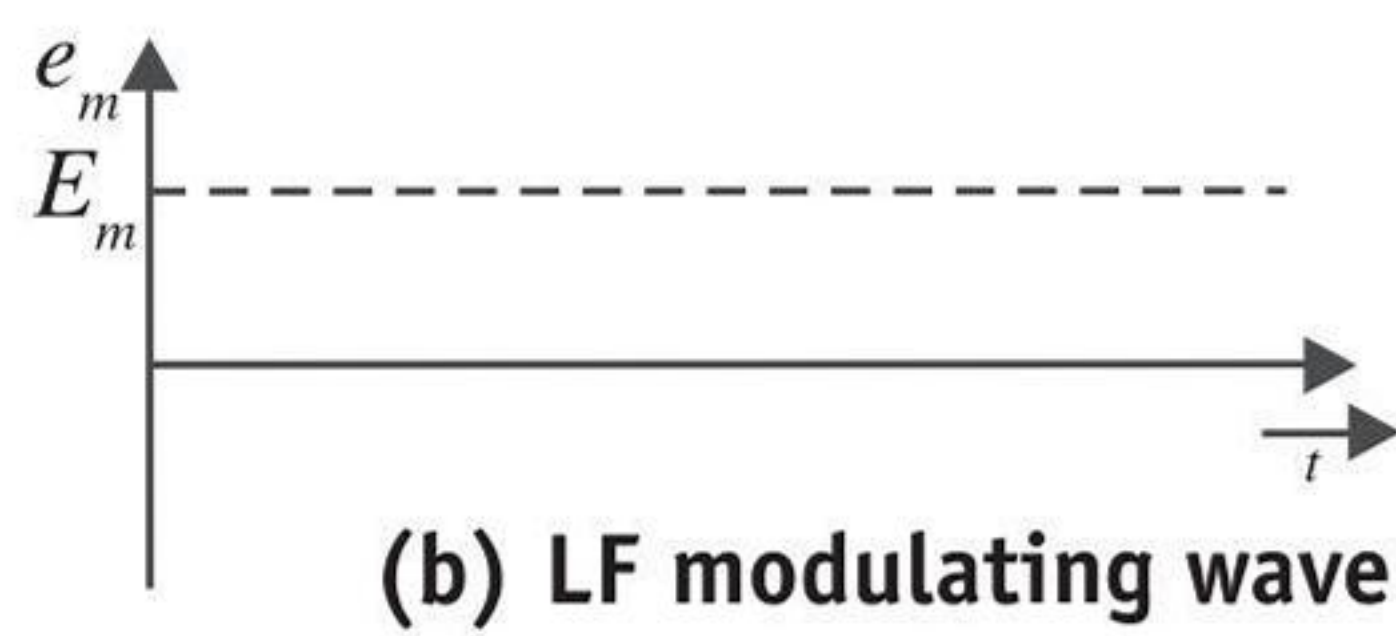
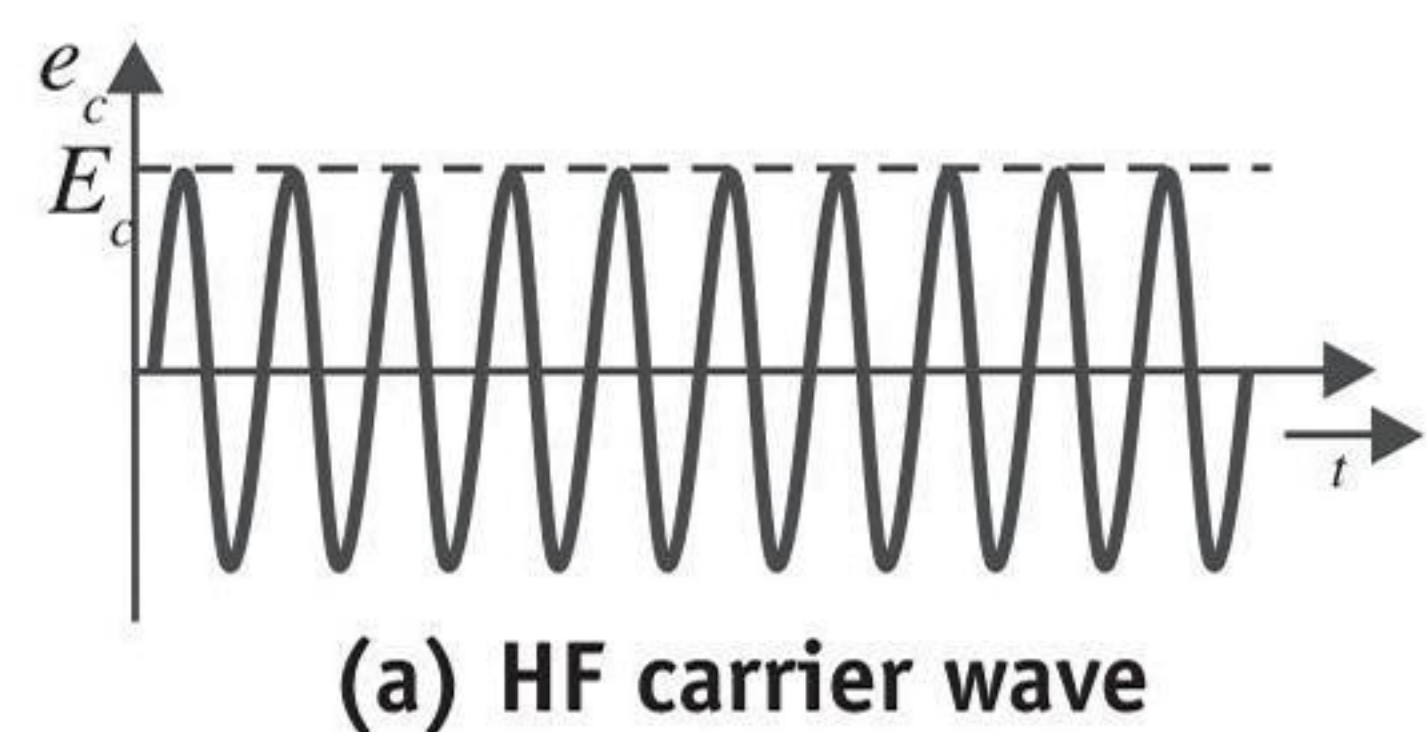
Limitations of Amplitude Modulation

1. Noisy reception
2. Low efficiency
3. Small operating range
4. Poor audio quality

FREQUENCY MODULATION

The process of changing the frequency of a carrier wave in accordance with the audio frequency signal is known as frequency modulation.

- Audio quality of AM transmission is poor. There is always a need to eliminate amplitude sensitive noise. This is possible if we eliminate amplitude variation. (i.e., a need to keep the amplitude of the carrier constant). This is precisely what we do in FM.
- In FM the overall amplitude of FM wave remains constant at all times.
- In FM, the total transmitted power remains constant.



Frequency deviation: The maximum change in frequency from mean value (f_c) is known as frequency deviation. This is also the change or shift either above or below the frequency f_c and is called as frequency deviation.

$$\therefore \delta = (f_{\max} - f_c) = f_c - f_{\min} = k_f \cdot \frac{E_m}{2\pi}$$

where k_f is the constant of proportionality. It determines the maximum variation in frequency of the modulated wave for a given modulating signal.

Carrier swing (CS): The total variation in frequency from the lowest to the highest is called the carrier swing, i.e.,

$$CS = 2 \times \Delta f$$

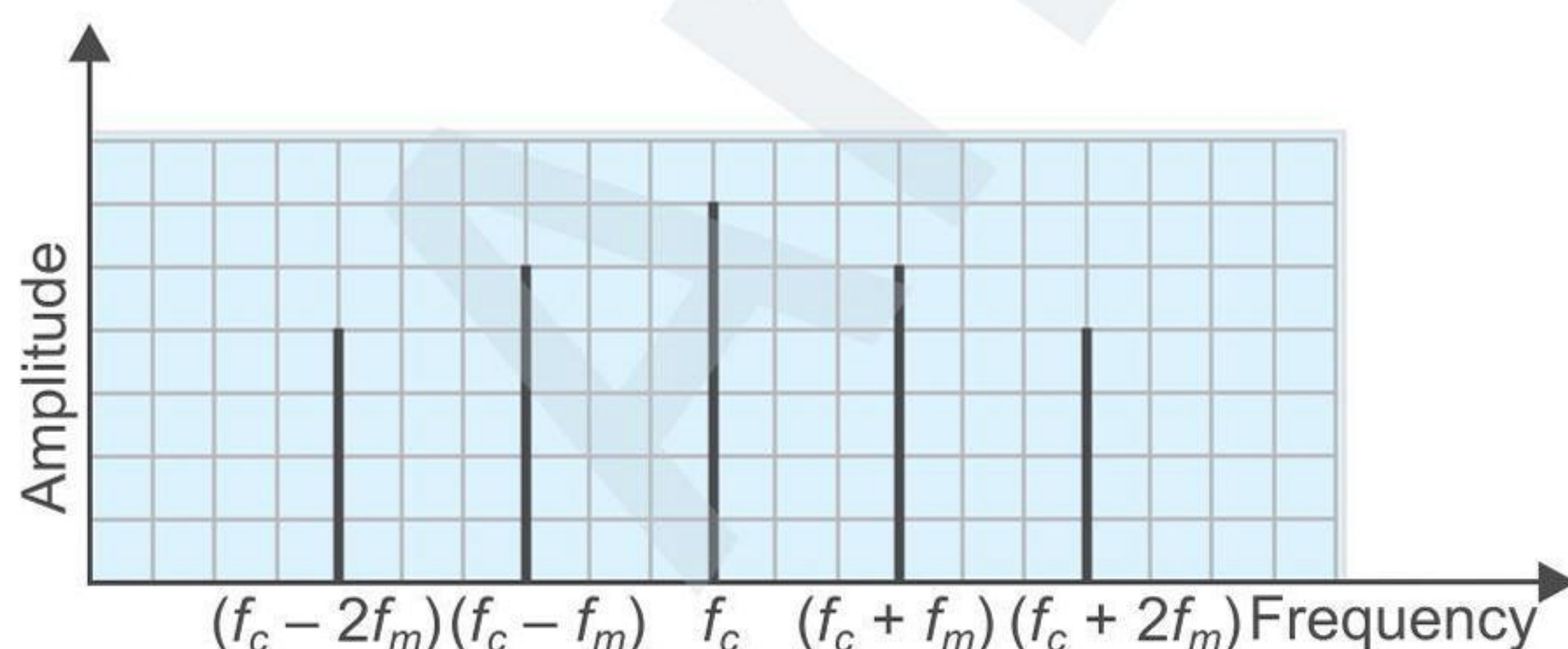
Frequency modulation index (m_f): The ratio of maximum frequency deviation to the modulating frequency is called modulation index.

$$m_f = \frac{\delta}{f_m} = \frac{f_{\max} - f_c}{f_m} = \frac{f_c - f_{\min}}{f_m} = \frac{k_f E_m}{f_m}$$

Frequency spectrum: FM side band modulated signal consist of infinite number of side bands whose frequencies are

$$(f_c \pm f_m), (f_c \pm 2f_m), (f_c \pm 3f_m), \dots$$

The number of side bands depends on the modulation index m_f .



In FM signal, the information (audio signal) is contained in the side bands. Since the side bands are separated from each other by the frequency of modulating signal f_m , so

$$\text{Bandwidth} = 2n \times f_m$$

where n = number of significant side band pairs

Deviation ratio: The ratio of maximum permitted frequency deviation to the maximum permitted audio frequency is known as deviation ratio. Thus,

$$\text{Deviation ratio} = \frac{(\Delta f)_{\max}}{(f_m)_{\max}}$$

PERCENT MODULATION

The ratio of actual frequency deviation to the maximum allowed frequency deviation is defined as percent modulation. Thus,

$$\text{Percent modulation, } m = \frac{(\Delta f)_{\text{actual}}}{(\Delta f)_{\max}}$$

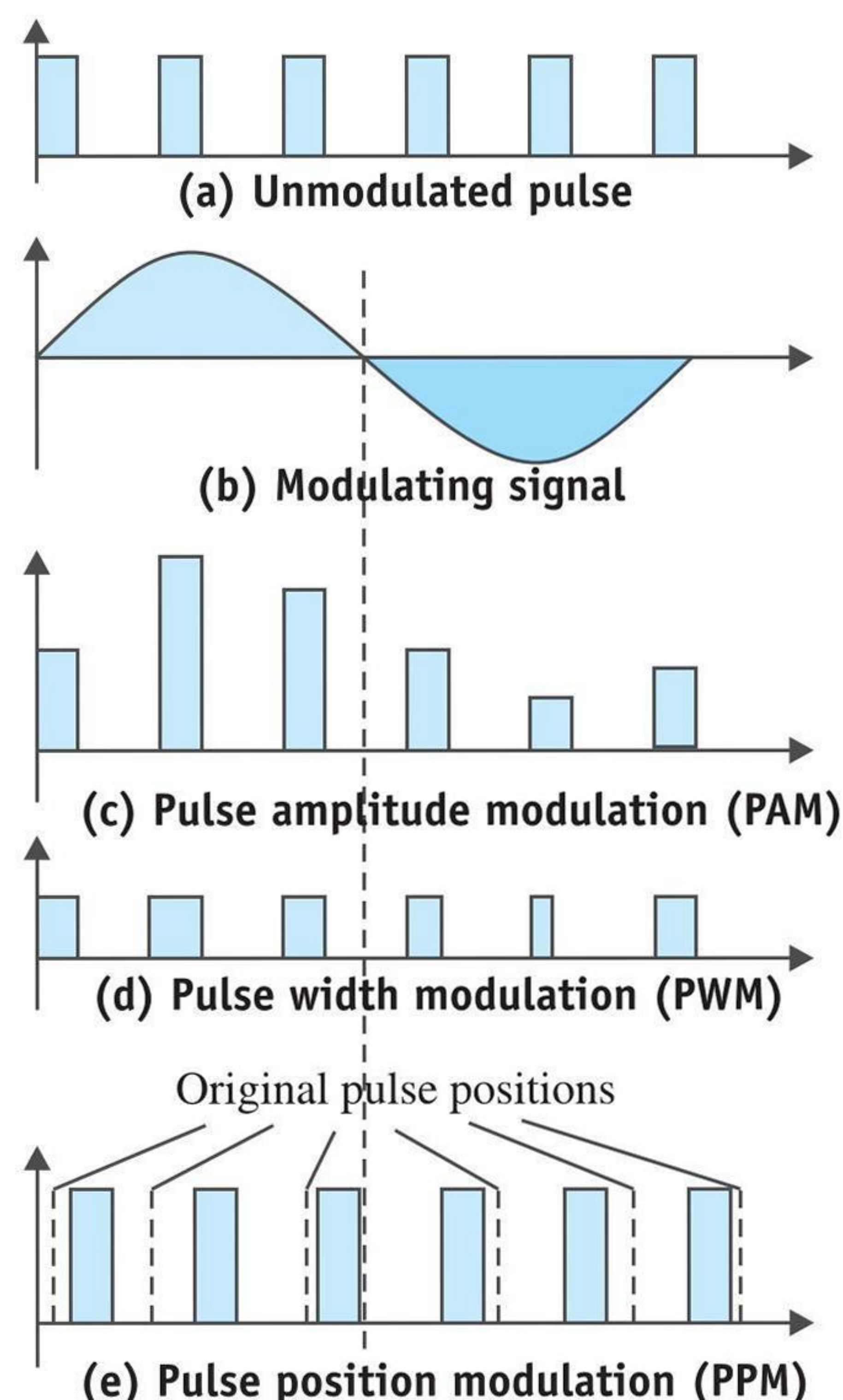
PULSE MODULATION

In pulse modulation, the carrier wave is in the form of pulses. Mentioned below are the types of pulse modulation.

Pulse amplitude modulation (PAM): The amplitude of the pulse varies in accordance with the modulating signal.

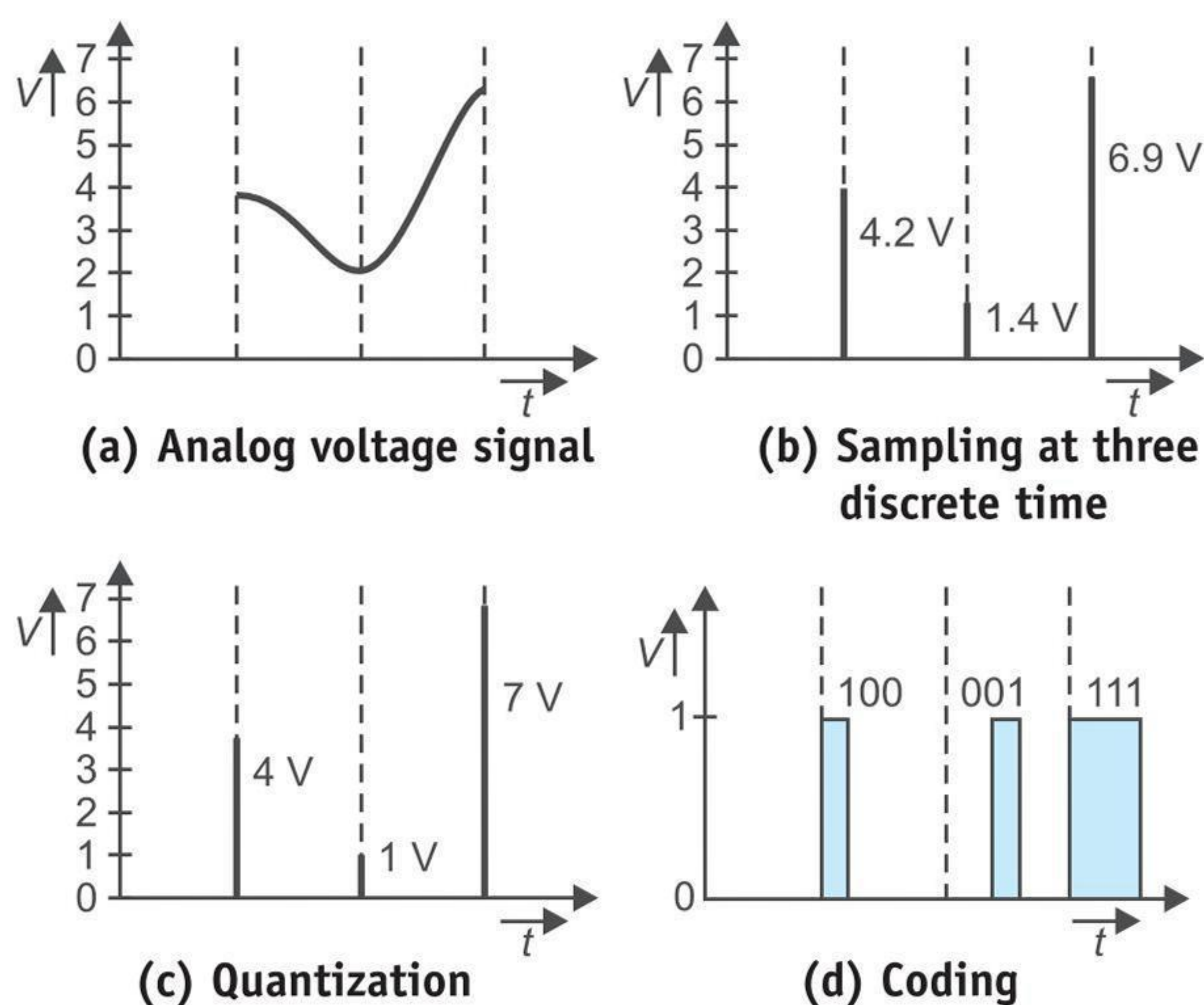
Pulse width modulation (PWM): The pulse duration varies in accordance with the modulating signal.

Pulse position modulation (PPM): In PPM, the position of the pulses of the carrier wave train is varied in accordance with the instantaneous value of the modulating signal.



PULSE CODE MODULATION

The pulse amplitude, pulse width, and pulse position modulations are not completely digital. A completely digital modulation is obtained by pulse code modulation (PCM). An analog signal is pulse code modulated by following three operations:



Sampling: It is the process of generating pulses of zero width and of amplitude equal to the instantaneous amplitude of the analog signal. The number of samples taken per second is called sampling rate.

Quantization: The process of dividing the maximum amplitude of the analog voltage signal into a fixed number of levels is called quantization. For example, amplitude 5 V of the analog voltage signal divides into six quantization levels, viz., 0, 1, 2, 3, 4, 5.

Pulses having amplitude between -0.5 V and 0.5 V are approximated (quantized) to a value of 0 V, amplitude between 0.5 V and 1.5 V are approximated to a value of 1 V, and so on.

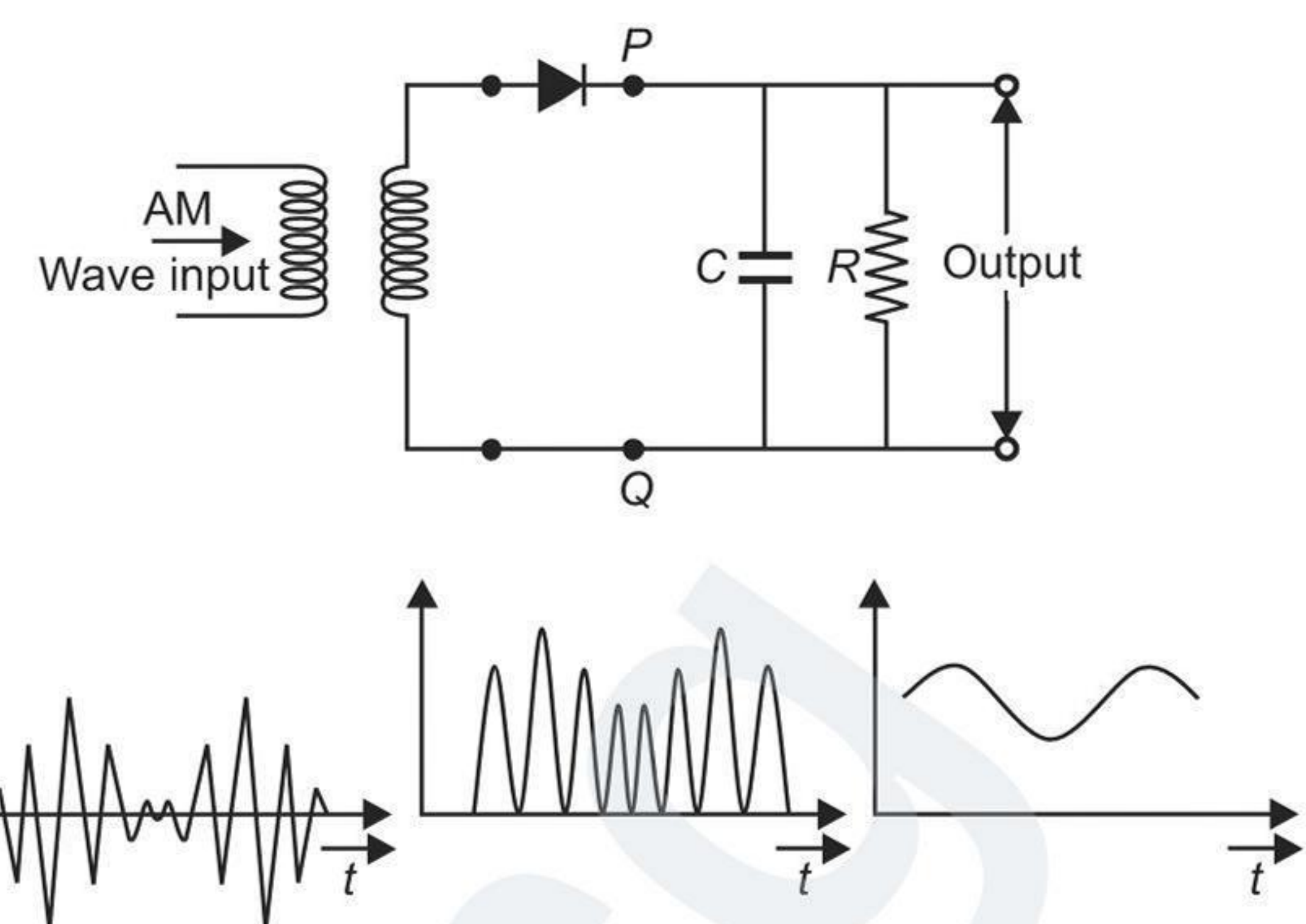
Coding: The process of digitizing the quantized pulses according to some code is called coding. For example, consider that voltage amplitude of an analog signal varies between 0 and 7 V.

DEMODULATION

The process of extracting the audio signal from the modulated wave is known as demodulation or detection. The wireless signals consist of radio frequency (high frequency) carrier wave modulated by audio frequency (low frequency). The diaphragm of a telephone receiver or a loud speaker cannot vibrate with high frequency. So it is necessary to separate the audio frequencies from the radio frequency carrier wave.

Simple demodulator circuit: A diode can be used to detect or demodulate an amplitude modulated (AM) wave. A diode basically acts as a rectifier, i.e., it reduces the modulated carrier wave into positive envelope only.

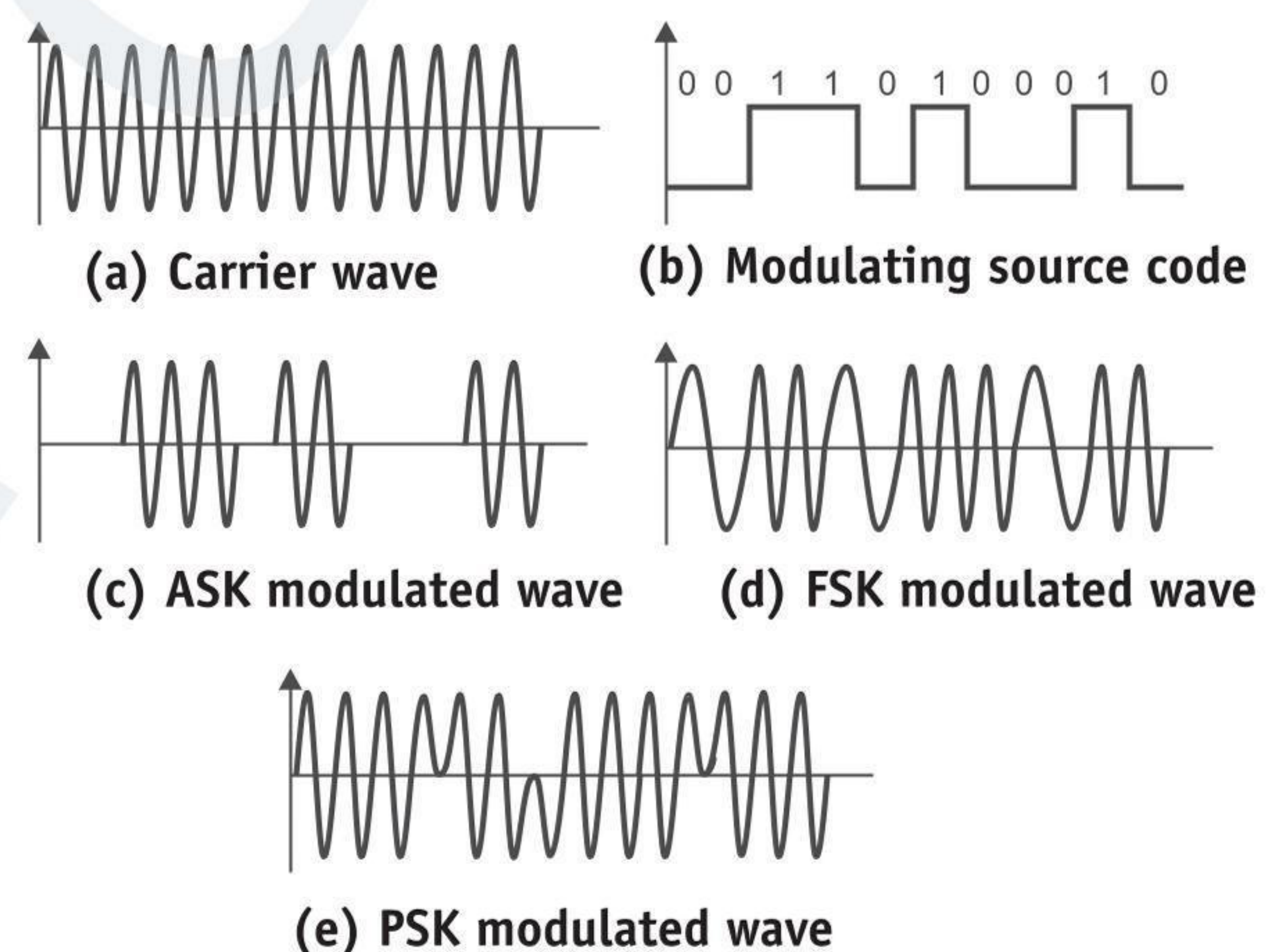
An AM wave input is shown in figure. It appears at the output of the diode across PQ as a rectified wave (since a diode conducts only in the positive half cycle). This rectified wave after passing through the RC network does not contain the radio frequency carrier component. Instead, it has only the envelope of the modulated wave.



In the actual circuit the value of RC is chosen such that $1/f_c \ll RC$, where f_c is the frequency of carrier signal.

DATA TRANSMISSION AND RETRIEVAL

The term data is applied to a representation of facts, concepts, or instructions suitable for communication, interpretation, or processing by human beings or by automatic means. Data in most cases consist of pulse type of signals.



The PCM signal is a series of 1s and 0s. Following three modulation techniques are used to transmit a PCM signal:

Amplitude shift keying (ASK): Two different amplitudes of the carrier represent the two binary values of the PCM signal. This method is also known as on-off keying (OOK):

- 1:** Presence of carrier of same constant amplitude.
- 0:** Carrier of zero amplitude.

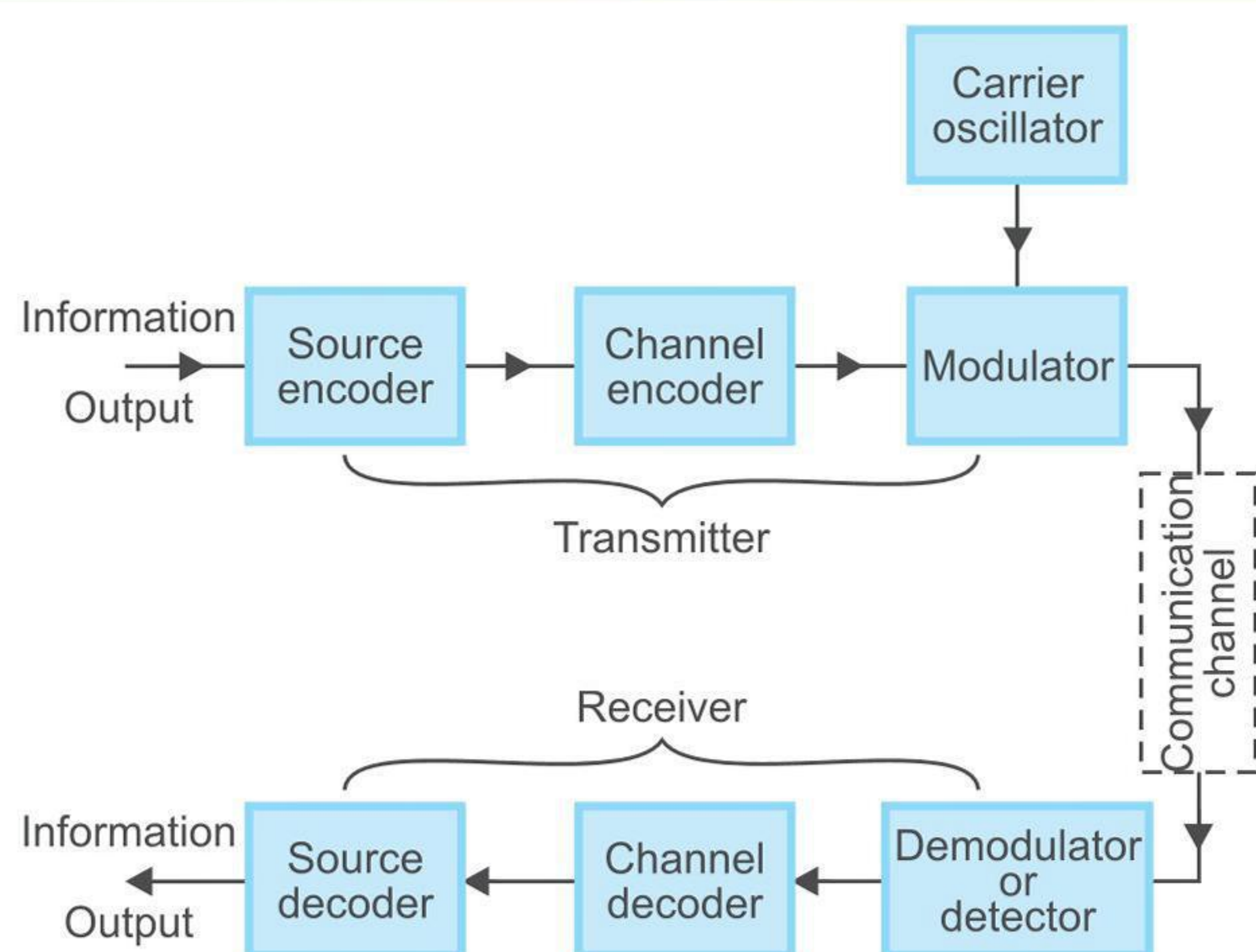
Frequency shift keying (FSK): The binary values of the PCM signal are represented by two frequencies.

- 1:** Increase in frequency.
- 0:** Frequency unaffected.

Phase shift keying (PSK): The phase of the carrier wave is changed in accordance with modulating data signal.

- 1:** Phase changed by π .
- 0:** Phase remains unchanged.

The analog signal is sampled by the sampler. The sampled pulses are then quantized. The encoder codes the quantized pulses according to the binary codes. After modulating the PCM signal (by ASK, FSK, or PSK method), the modulated signal is then transmitted into free space in the form of bits.

**ILLUSTRATION 9.1**

A TV transmitter has a range of 50 km. What is the height of the TV transmission tower? Radius of earth $R_e = 6.4 \times 10^6$ m.

Sol. Here $d = 50$ km $= 50 \times 10^3$ m

$\therefore$ Height of TV transmission tower

$$h = \frac{d^2}{2R_e} = \frac{(50 \times 10^3)^2}{2 \times 6.4 \times 10^6} = 195.3 \text{ m}$$

ILLUSTRATION 9.2

A TV tower has a height of 100 m. How much population is covered by the TV broadcast if the average population density around the tower is 1000 km^{-2} (radius of earth $= 6.4 \times 10^6$ m).

Sol. Here $h = 100$ m, $R_e = 6.4 \times 10^6$ m

Average population density

$$= 1000 \text{ km}^{-2} = 1000 \times (1000 \text{ m})^{-2} = 10^{-3} \text{ m}^{-2}$$

The TV transmission range, $d = \sqrt{2hR_e}$

Therefore, area of TV transmission range

$$\begin{aligned} A &= \pi d^2 = \pi 2h R_e = 2\pi h R_e \\ &= 2 \times 3.14 \times 100 \times 6.4 \times 10^6 \text{ m}^2 = 4.0 \times 10^9 \text{ m}^2 \end{aligned}$$

Therefore, population covered by TV broadcast

$$= 4.0 \times 10^9 \times 10^{-3} = 4.0 \times 10^6$$

ILLUSTRATION 9.3

On a particular day, the maximum frequency reflected from the ionosphere is 10 MHz. On another day, it was found to increase to 11 MHz. Calculate the ratio of maximum electron density of the ionosphere on the two days. Point out a plausible explanation for this.

Sol. $f_c = 9\sqrt{N_{\max}}$

$$N_{\max} = \frac{f_c^2}{81} \propto f_c^2 \Rightarrow \frac{(N_2)_{\max}}{(N_1)_{\max}} = \left(\frac{11}{10}\right)^2 = \frac{121}{100} = 1.21$$

The variation may be due to atmospheric disturbances.

ILLUSTRATION 9.4

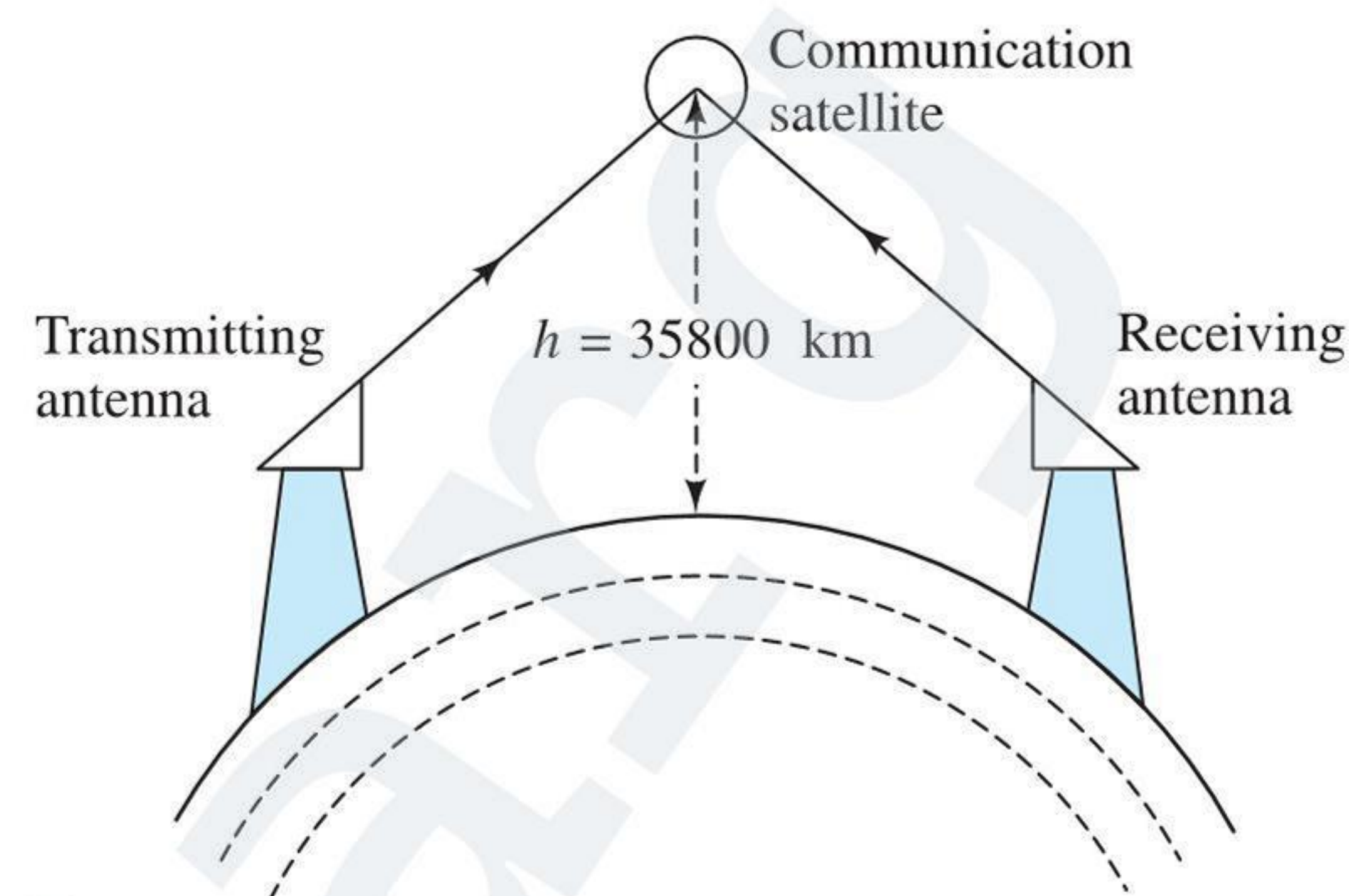
Frequencies higher than 10 MHz are found not to be reflected by the ionosphere on a particular day at a place. Calculate the maximum electron density of the ionosphere.

Sol. Given $f_c = 10 \text{ MHz} = 10 \times 10^6 \text{ Hz}$

If $N_{\max}$ is the maximum electron density, then $f_c = 9\sqrt{N_{\max}}$

$$N_{\max} = \frac{f_c^2}{81} = \frac{(10 \times 10^6)^2}{81} = 1.2 \times 10^{12} \text{ m}^{-3}$$

Satellite communication



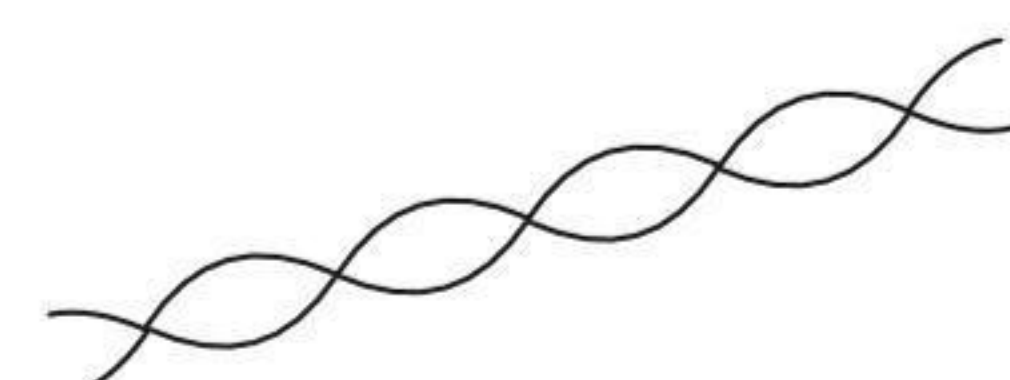
The height of communication satellite from earth's surface is 35,800 km. These communication satellites are required to cover the entire globe of earth.

REMOTE SENSING

Remote sensing is the process of finding information about an object at a distance without going near it. It is achieved by launching satellites in sun synchronous orbits.

LINE COMMUNICATION

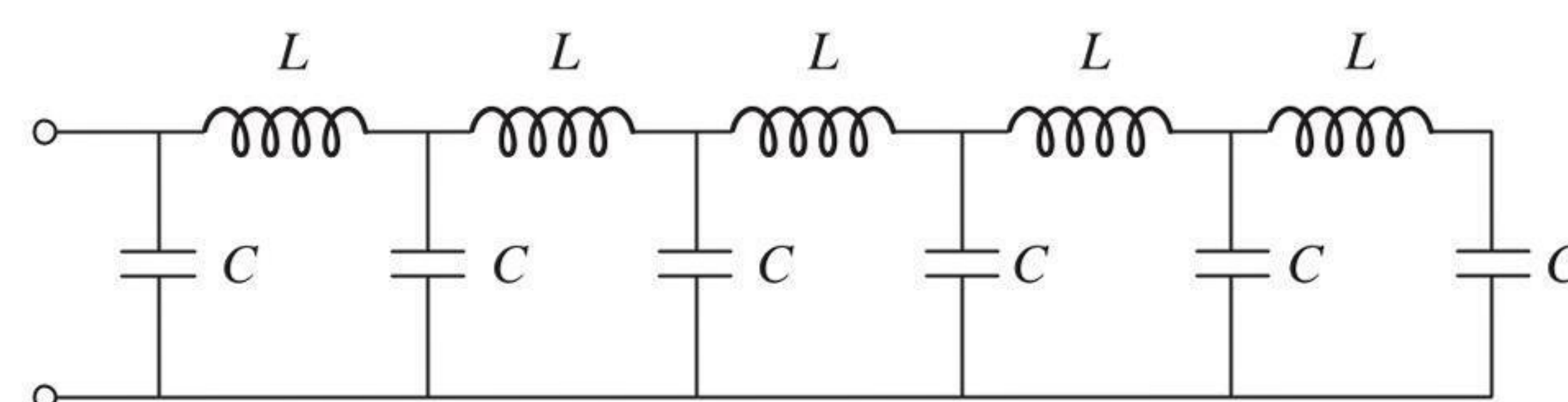
The line communication is conventional and it is used to transmit electricity and telephone signals through a pair of wires. The most used transmission lines are (i) a pair of conducting wires and (ii) a coaxial cable. It consists of two concentric conductors separated by an isolated material. The equivalent circuit of a transmission line consists of a sequence of inductance L and capacitance C .



(a) Twisted pair of wires



(b) Usual lines



(c) Equivalent circuit

OPTICAL COMMUNICATION

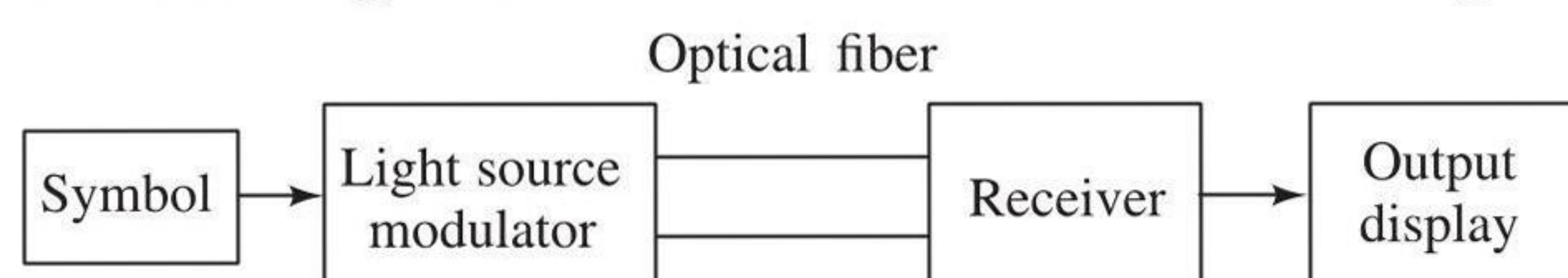
In optical communication, communication is through carrier optical signals. Optical signals cannot be transmitted through cables. The **essentials** of optical communication are

- Light source and modulator:** Light sources are LED, diode lasers.
- Communication channel:** The optical fibres of small diameters $= 10^{-5}$ m based on total internal reflection have been discovered.
- Optical signal detectors:** These are photodiode, Avalanche photodiode.

Advantages of optical communication are:

- (i) Wide channel band width has large channel carrying capacity.
- (ii) Low transmission losses
- (iii) Signal security

The schematic optical communication is shown in the figure.



LASER

Laser is abbreviation of light amplification of stimulated emission of radiation. It is based on stimulated emission. A laser device consists of a resonant cavity having one totally and one partially reflecting face. The active substance has metastable states and most of the atoms are in excited metastable state so that population inversion is achieved. When a photon of proper frequency is incident on the active material, it stimulates emission of photons. Finally an intense, monochromatic, highly directional coherent laser beam is available.

CONCEPT APPLICATION EXERCISE 9.1

- A sky wave with a frequency 55 MHz is incident on *D*-region of earth's atmosphere at 45° . The angle of refraction is (electron density for *D*-region is 400 electron/cm³)
 - (a) 60°
 - (b) 45°
 - (c) 30°
 - (d) 15°
- In a diode AM-detector, the output circuit consist of $R = 1 \text{ k}\Omega$ and $C = 10 \text{ pF}$. A carrier signal of 100 kHz is to be detected. Is it good?
 - (a) Yes
 - (b) No
 - (c) Information is not sufficient
 - (d) None of these
- A photodetector is made from a semiconductor $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ with $E_g = 0.73 \text{ eV}$. What is the maximum wavelength, which it can detect
 - (a) 1000 nm
 - (b) 1703 nm
 - (c) 500 nm
 - (d) 173 nm
- A transmitter supplies 9 kW to the aerial when unmodulated. The power radiated when modulated to 40% is
 - (a) 5 kW
 - (b) 9.72 kW
 - (c) 10 kW
 - (d) 12 kW
- The antenna current of an AM transmitter is 8 A when only carrier is sent but increases to 8.96 A when the carrier is sinusoidally modulated. The percentage modulation is
 - (a) 50%
 - (b) 60%
 - (c) 65%
 - (d) 71%
- The total power content of an AM wave is 1500 W. For 100% modulation, the power transmitted by the carrier is

- (a) 500 W
- (b) 700 W
- (c) 750 W
- (d) 1000 W

- The total power content of an AM wave is 900 W. For 100% modulation, the power transmitted by each side band is

- (a) 50 W
- (b) 100 W
- (c) 150 W
- (d) 200 W

- The modulation index of an FM carrier having a carrier swing of 200 kHz and a modulating signal 10 kHz is

- (a) 5
- (b) 10
- (c) 20
- (d) 25

- A 500-Hz modulating voltage fed into an FM generator produces a frequency deviation of 2.25 kHz. If amplitude of the voltage is kept constant but frequency is raised to 6 kHz, then the new deviation will be

- (a) 4.5 kHz
- (b) 54 kHz
- (c) 27 kHz
- (d) 15 kHz

- The bit rate for a signal, which has a sampling rate of 8 kHz and where 16 quantisation levels have been used is

- (a) 32000 bits/sec
- (b) 16000 bits/sec
- (c) 64000 bits/sec
- (d) 72000 bits/sec

- An amplitude modulated wave is modulated to 50%. What is the saving in power if carrier as well as one of the side bands are suppressed?

- (a) 70%
- (b) 65.4%
- (c) 94.4%
- (d) 25.5%

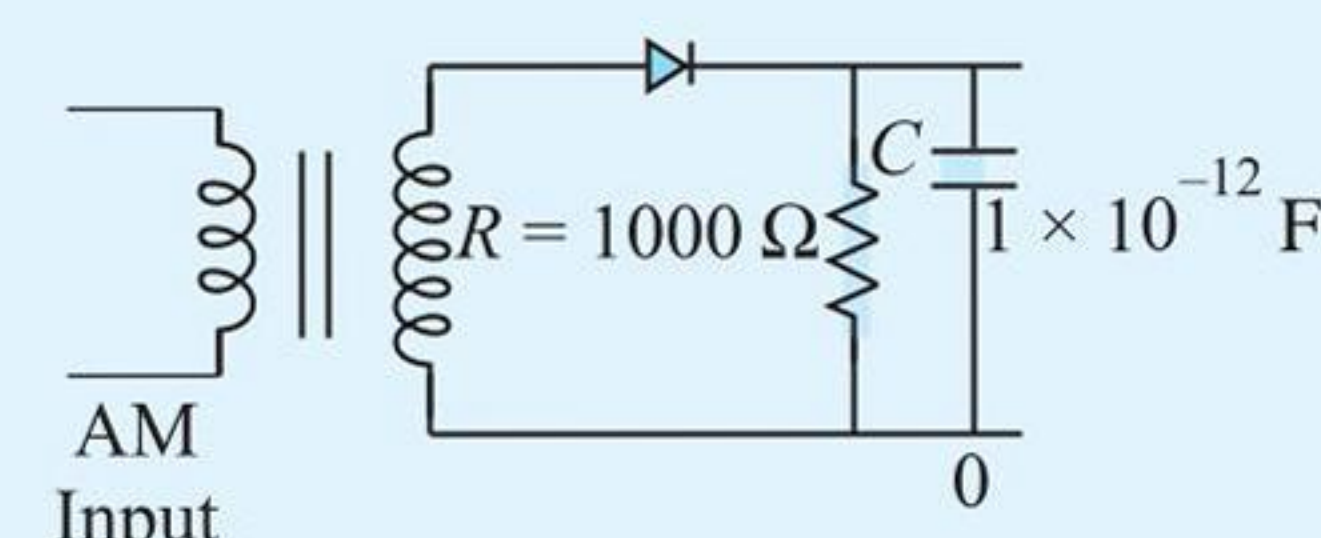
- In a radio receiver, the short wave and medium wave stations are tuned by using the same capacitor but coils of different inductance L_s and L_m , respectively, then

- (a) $L_s > L_m$
- (b) $L_s < L_m$
- (c) $L_s = L_m$
- (d) None of these

- A ground receiver station is receiving a signal at 5 MHz and transmitted from a ground transmitter at a height of 300 m, located at a distance of 100 km from the receiver station. The signal is coming via. [Radius of earth = $6.4 \times 10^6 \text{ m}$, N_{max} of isosphere = 10^{12} m^3]

- (a) Space wave
- (b) Sky wave propagation
- (c) Satellite transponder
- (d) All of these

- In the given detector circuit, the suitable value of carrier frequency is



- (a) $\ll 10^9 \text{ Hz}$
- (b) $\ll 10^5 \text{ Hz}$
- (c) $\gg 10^9 \text{ Hz}$
- (d) None of these

ANSWERS

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (b) | 4. (b) | 5. (d) |
| 6. (d) | 7. (c) | 8. (b) | 9. (b) | 10. (a) |
| 11. (c) | 12. (b) | 13. (b) | 14. (a) | |

Exercises

- In short wave communication waves, which of the following frequencies will be reflected back by the ionospheric layer, having electron density 10^{11} per m^3 ?
 (1) 2 MHz (2) 10 MHz
 (3) 12 MHz (4) 18 MHz
- Range of frequencies allotted for commercial FM radio broadcast is
 (1) 88 to 108 MHz (2) 88 to 108 kHz
 (3) 8 to 88 MHz (4) 88 to 108 GHz
- The velocity factor of a transmission line x . If dielectric constant of the medium is 2.6, the value of x is
 (1) 0.26 (2) 0.62
 (3) 2.6 (4) 6.2
- A step index fibre has a relative refractive index of 0.88%. What is the critical angle at the core-cladding interface?
 (1) 60° (2) 75°
 (3) 45° (4) None of these
- A laser beam of pulse power 10^{12} watt is focussed on an object area 10^{-4} cm^2 . The energy flux in watt/cm^2 at the point of focus is
 (1) 10^{20} (2) 10^{16}
 (3) 10^8 (4) 10^4
- The carrier frequency generated by a tank circuit containing 1 nF capacitor and 10 μH inductor is
 (1) 1592 Hz (2) 1592 MHz
 (3) 1592 kHz (4) 159.2 Hz
- For television broadcasting, the frequency employed is normally
 (1) 30–300 MHz (2) 30–300 GHz
 (3) 30–300 kHz (4) 30–300 Hz
- Maximum usable frequency (MUF) in F -region layer is x , when the critical frequency is 60 MHz and the angle of incidence is 70° . Then x is
 (1) 150 MHz (2) 170 MHz
 (3) 175 MHz (4) 190 MHz
- Laser beams are used to measure long distances because
 (1) they are monochromatic
 (2) they are highly polarised
 (3) they are coherent
 (4) they have high degree of parallelism
- An oscillator is producing FM waves of frequency 2 kHz with a variation of 10 kHz. What is the modulating index?
 (1) 0.20 (2) 5.0
 (3) 0.67 (4) 1.5
- The maximum peak to peak voltage of an AM wave is 24 mV and the minimum peak to peak voltage is 8 mV. The modulation factor is
 (1) 10% (2) 20%
 (3) 25% (4) 50%
- Sinusoidal carrier voltage of frequency 1.5 MHz and amplitude 50 V is amplitude modulated by sinusoidal voltage of frequency 10 kHz producing 50% modulation. The lower and upper side-band frequencies in kHz are
 (1) 1490, 1510 (2) 1510, 1490
 (3) $\frac{1}{1490}, \frac{1}{1510}$ (4) $\frac{1}{1510}, \frac{1}{1490}$
- Which of the following is the disadvantage of FM over AM?
 (1) Larger band width requirement
 (2) Larger noise
 (3) Higher modulation power
 (4) Low efficiency
- An AM wave has 1800 W of total power content. For 100% modulation, the carrier should have power content equal to
 (1) 1000 watt (2) 1200 watt
 (3) 1500 watt (4) 1600 watt
- An antenna is a device
 (1) that converts electromagnetic energy into radio frequency signal
 (2) that converts radio frequency signal into electromagnetic energy
 (3) that converts guided electromagnetic waves into free space electromagnetic waves and vice-versa
 (4) none of these
- In an FM system, a 7 kHz signal modulates 108 MHz carrier so that frequency deviation is 50 kHz. The carrier swing is
 (1) 7.143 (2) 8
 (3) 0.71 (4) 350
- The phenomenon by which light travels in an optical fibre is
 (1) reflection (2) refraction
 (3) total internal reflection (4) transmission
- Advantage of optical fibre
 (1) high bandwidth and no interference
 (2) low bandwidth and no interference
 (3) high bandwidth, low transmission capacity and no e.m. interference
 (4) high bandwidth, high data transmission capacity and no e.m. interference
- In frequency modulation
 (1) the amplitude of modulated wave varies as frequency of carrier wave
 (2) the frequency of modulated wave varies as amplitude of modulating wave
 (3) the amplitude of modulated wave varies as amplitude of carrier wave
 (4) the frequency of modulated wave varies as frequency of modulating wave
- Audio signal cannot be transmitted because
 (1) the signal has more noise
 (2) the signal cannot be amplified for distance communication
 (3) the transmitting antenna length is very small to design
 (4) the transmitting antenna length is very large and impracticable
- In which of the following remote sensing technique is not used?
 (1) Forest density (2) Pollution
 (3) Wetland mapping (4) Medical treatment

22. For sky wave propagation of a 10 MHz signal, what should be the minimum electron density in ionosphere?
 (1) $\sim 1.2 \times 10^{12} \text{ m}^{-3}$ (2) $\sim 10^6 \text{ m}^{-3}$
 (3) $\sim 10^{14} \text{ m}^{-3}$ (4) $\sim 10^{22} \text{ m}^{-3}$
23. What should be the maximum acceptance angle at the aircore interface of an optical fibre if n_1 and n_2 are the refractive indices of the core and the cladding, respectively?
 (1) $\sin^{-1}(n_2/n_1)$ (2) $\sin^{-1}\sqrt{n_1^2 - n_2^2}$
 (3) $\left[\tan^{-1} \frac{n_2}{n_1} \right]$ (4) $\left[\tan^{-1} \frac{n_1}{n_2} \right]$
24. The audio signal used to modulate $60 \sin(2\pi \times 10^6 t)$ is $15 \sin 300 \pi t$. The depth of modulation is
 (1) 50% (2) 40%
 (3) 25% (4) 15%
25. The impedance of coaxial cable, when its inductance is $0.40 \mu\text{H}$ and capacitance is $1 \times 10^{-11} \text{ F}$, can be
 (1) $2 \times 10^2 \Omega$ (2) 100Ω
 (3) $3 \times 10^3 \Omega$ (4) $3 \times 10^{-2} \Omega$
26. An optical fibre communication system works on a wavelength of $1.3 \mu\text{m}$. The number of subscribers it can feed if a channel requires 20 kHz are
 (1) 2.3×10^{10} (2) 1.15×10^{10}
 (3) 1×10^5 (4) None of these
27. In an FM system a 7 kHz signal modulates 108 MHz carrier so that frequency deviation is 50 kHz. The carrier swing is
 (1) 7.143 (2) 8
 (3) 0.71 (4) 350
28. The electron density of E , F_1 , F_2 layers of ionosphere is 2×10^{11} , 5×10^{11} and $8 \times 10^{11} \text{ m}^{-3}$ respectively. What is the ratio of critical frequency for reflection of radiowaves?
 (1) 2 : 4 : 3 (2) 4 : 3 : 2
 (3) 2 : 3 : 4 (4) 3 : 2 : 4
29. Mean optical power launched into an 8 km fibre is $120 \mu\text{W}$ and mean output power is $4 \mu\text{W}$, then the overall attenuation is (Given $\log 30 = 1.477$)
 (1) 14.77 dB (2) 16.77 dB
 (3) 3.01 dB (4) None of these
30. An antenna current of an AM broadcast transmitter modulated by 50% is 11 A. The carrier current is
 (1) 10.35 A (2) 9.25 A
 (3) 10 A (4) 5.5 A
31. A transmitter transmits a power of 10 kW when modulation is 50%. Power of carrier wave is
 (1) 5 kW (2) 8.89 kW
 (3) 14 kW (4) 5.7 kW
32. A telephone link operating at a central frequency of 10 GHz is established. If 1% of this is available then how many telephone channel can be simultaneously given when each telephone covering a band width of 5 kHz
 (1) 2×10^4 (2) 2×10^6
 (3) 5×10^4 (4) 5×10^6
33. In AM, the cent percent modulation is achieved when
 (1) carrier amplitude = signal amplitude
 (2) carrier amplitude $\neq$ signal amplitude
 (3) carrier frequency = signal frequency
 (4) carrier frequency $\neq$ signal frequency

Answers Key

EXERCISES

Single Correct Answer Type

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (1) | 2. (1) | 3. (2) | 4. (4) | 5. (2) |
| 6. (3) | 7. (1) | 8. (3) | 9. (4) | 10. (2) |
| 11. (4) | 12. (1) | 13. (1) | 14. (2) | 15. (3) |

- | | | | | |
|---------|---------|---------|---------|---------|
| 16. (1) | 17. (3) | 18. (4) | 19. (4) | 20. (4) |
| 21. (4) | 22. (1) | 23. (2) | 24. (3) | 25. (1) |
| 26. (2) | 27. (1) | 28. (3) | 29. (1) | 30. (1) |
| 31. (2) | 32. (1) | 33. (1) | | |

Chapter 9

Concept Application Exercises

Exercise 9.1

$$1. (b) n_{eff} = n_0 \sqrt{1 - \left(\frac{80.5 N}{v^2} \right)} = 1 \sqrt{1 - \frac{80.5 \times (400 \times 10^6)}{(55 \times 10^6)^2}} \approx 1$$

$$\text{Also } n_{eff} = \frac{\sin i}{\sin r}$$

$$\Rightarrow \sin r = \sin i \Rightarrow r = i = 45^\circ$$

$$2. (b) \text{ For demodulation, } \frac{1}{f_c} \ll RC$$

$$\frac{1}{f_c} = \frac{1}{100 \times 10^3} = 10^{-5} \text{ s}$$

$$RC = 10^3 \times 10 \times 10^{-12} \text{ s} = 10^{-8} \text{ s}$$

We see that $1/f_c$ here is not less than RC as required by the above condition. Hence, this is not good.

$$3. (b) \text{ Limiting value of } h\nu \text{ is } E_g, \text{ such that } h\nu = \frac{hc}{\lambda} = E_g$$

$$\text{or } \lambda = \frac{hc}{E_g} = \frac{6.63 \times 10^{-34} \text{ J-s} \times 3 \times 10^8 \text{ ms}^{-1}}{0.73 \times 1.6 \times 10^{-19} \text{ J}} = 1703 \text{ nm}$$

$$4. (b) P_t = P_c \left[1 + \frac{m^2}{2} \right] = 9 \left[1 + \frac{(0.4)^2}{2} \right] = 9 \left[1 + \frac{0.16}{2} \right]$$

$$(\because m = 40\% = 0.4)$$

$$= 9 (1.08) = 9.72 \text{ kW}$$

$$5. (d) \text{ We know that } \left(\frac{I_t}{I_c} \right)^2 = 1 + \frac{m^2}{2}$$

$$\text{Here, } I_t = 8.96 \text{ A and } I_c = 8 \text{ A}$$

$$\therefore \left(\frac{8.96}{8} \right)^2 = 1 + \frac{m^2}{2}$$

$$\text{or } 1.254 = 1 + \frac{m^2}{2}$$

$$\text{or } \frac{m^2}{2} = 0.254$$

$$\text{or } m^2 = 0.508 \quad \text{or } m = 0.71 = 71\%$$

$$6. (d) \frac{P_t}{P_c} = 1 + \frac{m^2}{2} \quad \text{or } P_c = P_t \left[\frac{2}{2 + m^2} \right]$$

$$\therefore P_c = 1500 \left[\frac{2}{2 + 1} \right] = 1000 \text{ W}$$

$$\therefore m = 100\% = 1$$

$$7. (c) P_c = P_t \left[\frac{2}{2 + m^2} \right] = 900 \left[\frac{2}{2 + 1} \right] = 600 \text{ W}$$

$$\text{Now, } P_{LSB} = \frac{m^2}{4} \times P_c = \frac{1}{4} \times 600 = 150 \text{ W}$$

$$8. (b) CS = 2 \times \Delta f \text{ or } \Delta f = CS/2$$

$$\therefore \Delta f = \frac{200}{2} = 100 \text{ kHz}$$

$$\text{Now } m_f = \frac{\Delta f}{f_m} = \frac{100}{10} = 10$$

$$9. (b) m_f = \frac{\delta}{f_m} = \frac{2250}{500} = 4.5$$

$$\therefore \text{ New deviation} = 2(m_f f_m) = 2 \times 4.5 \times 6 = 54 \text{ kHz}$$

$$10. (a) \text{ If } n \text{ is the number of bits per sample, then number of quantisation level} = 2^n$$

$$\text{Since the number of quantisation level is 16}$$

$$\Rightarrow 2^n = 16 \Rightarrow n = 4$$

$$\therefore \text{ Bit rate} = \text{Sampling rate} \times \text{No. of bits per sample}$$

$$= 8000 \times 8 = 32000 \text{ bits/sec}$$

$$11. (c) P_{sb} = P_c \left(\frac{m_a}{2} \right)^2 = P_c \frac{(0.5)^2}{4} = 0.0625 P_c$$

$$\text{Also } P = P_c \left(1 + \frac{m_a^2}{2} \right) = P_c \left(1 + \frac{(0.5)^2}{2} \right) = 1.125 P_c$$

$$\therefore \% \text{ saving} = \frac{(1.125 P_c - 0.0625 P_c)}{1.125 P_c} \times 100 = 94.4\%$$

$$12. (b) \text{ As } f = \frac{c}{\lambda} \Rightarrow f_m = \frac{c}{\lambda_m} \text{ and } f_s = \frac{c}{\lambda_s}$$

$$\therefore \lambda_m > \lambda_s \Rightarrow f_m < f_s$$

$$\text{Also } f_m = \frac{1}{2\pi\sqrt{L_m C}} \text{ and } f_s = \frac{1}{2\pi\sqrt{L_s C}}$$

$$13. (b) \text{ Maximum distance covered by space wave communication}$$

$$\sqrt{2Rh} = 62 \text{ km}$$

$$\text{Critical frequency} = f_c = 9(N_{\max})^{1/2} \approx 9 \text{ MHz}$$

$$5 \text{ MHz} < f_c, \text{ sky wave propagation (ionospheric propagation)}$$

$$14. (a) \text{ Using } \frac{1}{f_{\text{carrier}}} \ll RC$$

$$\text{We get time constant, } RC = 1000 \times 10^{-12} = 10^{-9} \text{ s}$$

$$\text{Now } f = \frac{1}{T} = \frac{1}{10^{-9}} = 10^9 \text{ Hz}$$

Thus, the value of carrier frequency should be much less than 10^9 Hz , say 100 kHz .

$$\Rightarrow \frac{f_m}{f_s} = \sqrt{\frac{L_s}{L_m}} \Rightarrow L_s < L_m$$

Exercises

Single Correct Answer Type

$$1. (1) \text{ By using } f_c \approx 9(N_{\max})^{1/2} \Rightarrow f_c \approx 2 \text{ MHz}$$

2. (1) A maximum frequency deviation of 75 kHz is permitted for commercial FM broadcast stations in the 88 to 108 MHz VHF band.

$$3. (2) \text{ Velocity factor} = \frac{1}{\sqrt{k}} = \frac{1}{\sqrt{2.6}} = 0.62$$

$$4. (4) \text{ Here } \frac{n_1 - n_2}{n_1} = \frac{0.88}{100} \Rightarrow \frac{n_2}{n_1} = 0.9912$$

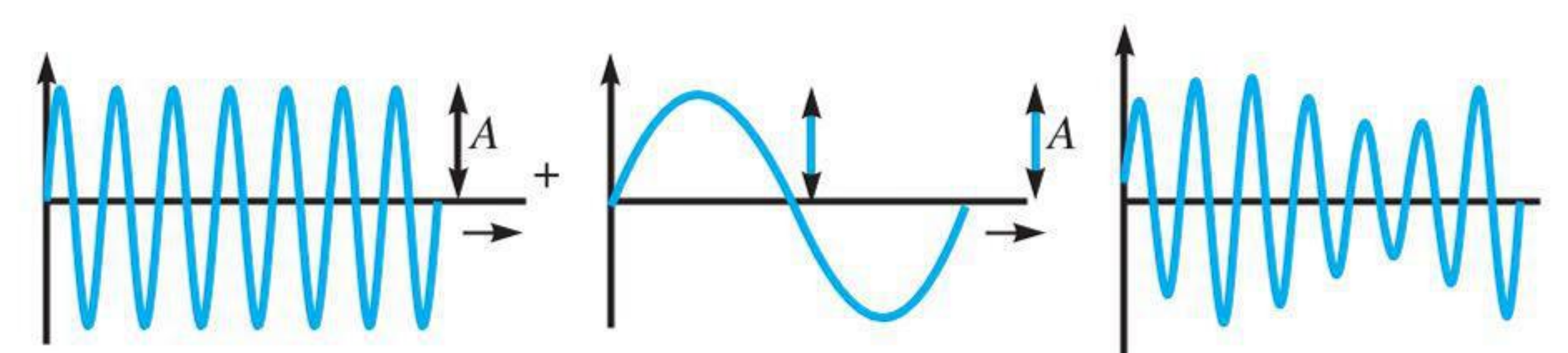
$$\therefore \text{ Critical angle, } \theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) = \sin^{-1}(0.9912) = 84^\circ 24'$$

5. (2) The energy flux, $\phi = \frac{\text{Pulse power}}{\text{Area}} = \frac{10^{12}}{10^{-4}} = 10^{16} \frac{\text{W}}{\text{cm}^2}$
6. (3) $\nu = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2 \times 3.14 \sqrt{10 \times 10^{-6} \times 1 \times 10^{-9}}} = 1592 \text{ kHz}$
7. (1) VHF (very high frequency) band having frequency range 30 MHz to 300 MHz is typically used for TV and radar transmission.
8. (3) $\text{MUF} = \frac{f_c}{\cos \theta} = \frac{60}{\cos 70^\circ} = 175 \text{ MHz}$
9. (4) Laser beams are perfectly parallel. So that they are very narrow and can travel a long distance without spreading. This is the feature of laser while they are monochromatic and coherent these are characteristics only.
10. (2) The formula for modulating index is given by

$$m_f = \frac{\delta}{v_m} = \frac{\text{Frequency variation}}{\text{Modulating frequency}} = \frac{10 \times 10^3}{2 \times 10^3} = 5$$
11. (4) Here, $V_{\max} = \frac{24}{2} = 12 \text{ mV}$ and $V_{\min} = \frac{8}{2} = 2 \text{ mV}$
 Now, $m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = \frac{12 - 4}{12 + 4} = \frac{8}{16} = \frac{1}{2} = 0.5 = 50\%$
12. (1) Here, $f_c = 1.5 \text{ MHz} = 1500 \text{ kHz}$, $f_m = 10 \text{ kHz}$
 $\therefore$ Low side band frequency
 $= f_c - f_m = 1500 \text{ kHz} - 10 \text{ kHz} = 1490 \text{ kHz}$
 Upper side band frequency
 $= f_c + f_m = 1500 \text{ kHz} + 10 \text{ kHz} = 1510 \text{ kHz}$
13. (1) Frequency modulation requires much wider channel (7 to 15 times) as compared to AM.
14. (2) $P_t = P_c \left(1 + \frac{m_a^2}{2}\right)$; Here $m_a = 1$
 $\Rightarrow 1800 = P_c \left(1 + \frac{(1)^2}{2}\right) \Rightarrow P_c = 1200 \text{ W}$
15. (3) An antenna is a metallic structure used to radiate or receive EM waves.
16. (1) Carrier swing $= \frac{\text{Frequency deviation}}{\text{Modulating frequency}} = \frac{50}{7} = 7.143$
17. (3) In optical fibre, light travels inside it, due to total internal reflection.
18. (4) Few advantages of optical fibres are that the number of signals carried by optical fibres is much more than that carried by the Cu wire or radio waves. Optical fibres are practically free from electromagnetic interference and problem of cross talks whereas ordinary cables and microwave links suffer a lot from it.
19. (4) The process of changing the frequency of a carrier wave (modulated wave) in accordance with the audio frequency signal (modulating wave) is known as frequency modulation (FM).
20. (4) Following are the problems which are faced while transmitting audio signals directly.
 (i) These signals are relatively of short range.
 (ii) If everybody started transmitting these low frequency signals directly, mutual interference will render all of them ineffective.

(iii) Size of antenna required for their efficient radiation would be larger i.e. about 75 km.

21. (4) Remote sensing is the technique to collect information about an object in respect of its size, colour, nature, location, temperature etc. without physically touching it. There are some areas or location which are inaccessible. So to explore these areas or locations, a technique known as remote sensing is used. Remote sensing is done through a satellite.
22. (1) The critical frequency of a sky wave for reflection from a layer of atmosphere is given by $f_c = 9(N_{\max})^{1/2}$
 $\Rightarrow 10 \times 10^6 = 9(N_{\max})^{1/2}$
 $\Rightarrow N_{\max} = \left(\frac{10 \times 10^6}{9}\right)^2 \approx 1.2 \times 10^{12} \text{ m}^{-3}$
23. (2) Core of acceptance angle, $\theta = \sin^{-1} \sqrt{n_1^2 - n_2^2}$
24. (3) $m_a = \frac{E_m}{E_c} = \frac{15}{60} \times 100 = 25\%$
25. (1) Using $Z = \sqrt{\frac{L}{C}}$, we get $Z = \sqrt{\frac{0.40 \times 10^{-6}}{10^{-11}}} = 2 \times 10^2 \Omega$
26. (2) Optical source frequency, $f = \frac{c}{\lambda} = \frac{3 \times 10^8}{1.3 \times 10^{-6}} = 2.3 \times 10^{14} \text{ Hz}$
 $\therefore$ Number of channels or subscribers $= \frac{2.3 \times 10^{14}}{20 \times 10^3} = 1.15 \times 10^{10}$
27. (1) Carrier swing $= \frac{\text{Frequency deviation}}{\text{Modulating frequency}} = \frac{50}{7} = 7.143$
28. (3) $f_c \propto (N)^{1/2}$
 $\Rightarrow (f)_{cE} : (f)_{cF1} : (f)_{cF2}$
 $= (2 \times 10^{11})^{1/2} : (5 \times 10^{11})^{1/2} : (8 \times 10^{11})^{1/2}$
 $= 2 : 3 : 4$
29. (1) Attenuation $= 10 \log \frac{120}{4} = 10 \log 30 = 10 \times 1.4771 = 14.77 \text{ dB}$
30. (1) $I_{\text{Carrier}} = \frac{I_{\text{rms}}}{\sqrt{1 + \frac{m_a^2}{2}}} = \frac{11}{\sqrt{1 + \frac{(0.5)^2}{2}}} = 10.35 \text{ A}$
31. (2) $P_c = \frac{P}{\left(1 + \frac{m_a^2}{2}\right)} = \frac{10000}{\left(1 + \frac{(0.5)^2}{2}\right)} = \frac{10000}{1.125} = 8.89 \text{ kW}$
32. (1) $1\% \text{ of } 10 \text{ GHz} = 10 \times 10^9 \times \frac{1}{100} = 10^8 \text{ Hz}$
 Number of channels $= \frac{10^8}{5 \times 10^3} = 2 \times 10^4$
33. (1) When signal amplitude is equal to the carrier amplitude, the amplitude of carrier wave varies between $2A$ and zero.



$$m_a = \frac{\text{Amplitude change of carrier}}{\text{Amplitude of normal carrier}} = \frac{2A - A}{A} \times 100 = 100\%$$

